

推理模型中的动态早期退出

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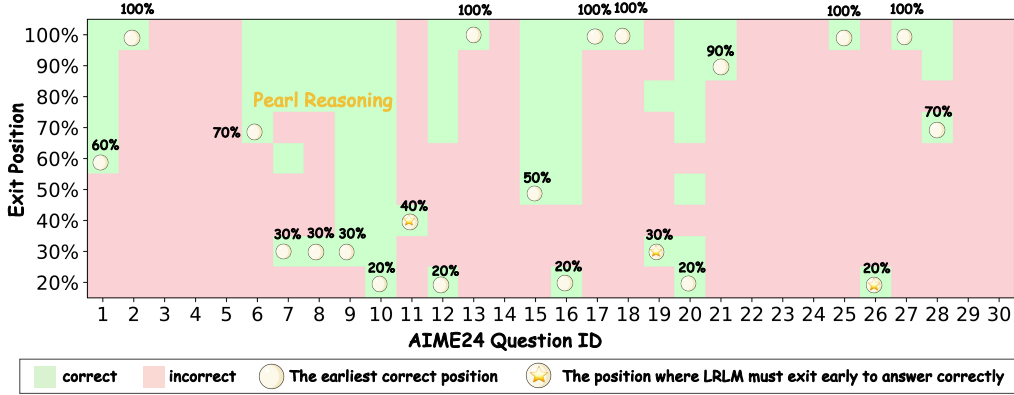


图 1: 不同推理步骤早期退出的正确率统计。

Abstract

大型推理语言模型（LRMs）的最新进展依赖于测试时扩展，即生成长链思维（CoT）来解决复杂任务。然而，长链思维中的过度思考不仅会降低问题解决的效率，而且由于极其详细或冗余的推理步骤，还存在准确性损失的风险。我们提出了一种简单而有效的方法，允许 LLMs 在生成过程中通过提前退出来自我截断 CoT 序列。该方法不依赖固定的启发式规则，而是在潜在的推理转换点监控模型行为，当模型对试探性答案表现出高置信度时，动态终止下一个推理链的生成。我们的方法无需额外训练，可以无缝集成到现有的类 o1 推理 LLMs 中。在 10 个推理基准测试（例如 GSM8K、MATH-500、AMC、GPQA、AIME 和 LiveCodeBench）上的实验表明，所提出的方法在 11 个不同系列和规模的尖端推理 LLMs 上持续有效，将 CoT 序列的长度平均减少了 19.1% 至 80.1%，同时将准确性提高了 0.3% 至 5.0%。¹

1 Introduction

大型推理模型（Large Reasoning Models, LRMs）的出现 (Xu et al., 2025a)，例如 DeepSeek-R1 (DeepSeek-AI et al., 2025) 和 GPT-O1 (OpenAI, 2025)，标志着自然语言处理领域，特别是在解决复杂精细任务方面 (WANG et al., 2025)，取得了重大突破。这些模型通过生成

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¹ 代码可在 <https://github.com/iie-ycx/DEER> 获取。

更长、包含丰富多样推理路径的思维链 (CoT) (Wei et al., 2023), 利用测试时缩放 (Snell et al., 2024) 定律, 释放了其推理能力的潜力。

然而, 生成长度过长的思维链会显著增加计算负荷和推理延迟, 这阻碍了它们在计算敏感型应用中的部署。此外, 最近的研究 (Chen et al., 2025b; Team et al., 2025a) 揭示了大型推理模型存在一个固有的过度思考问题: 这些模型会持续生成冗长的推理序列 (Wu et al., 2025; Cuadron et al., 2025), 引入无关信息和不必要的思考步骤。这种冗余处理不仅浪费计算资源, 还会导致准确率下降, 因为它使模型偏离正确的推理路径, 转向错误的路径 (参见图1中的问题 11、19 和 26)。这种冗余可归因于监督微调 (Achiam et al., 2023; Wei et al., 2021; Ouyang et al., 2022) 或强化学习 (Bai et al., 2022; Ouyang et al., 2022; Schulman et al., 2017; Ramesh et al., 2024) 阶段的设计, 其中忽略了在生成过程中动态调整其推理长度的能力, 从而在大型推理模型的推理效率上留下了空白。

直观上, 随着推理路径数量的增加, 生成结论时参考的信息也越多。如果我们能识别出推理信息恰好足够的关键点 (称为 Pearl Reasoning), 并迫使模型在此点停止进一步思考, 直接输出结论, 那么我们就同时实现准确性和效率。本文旨在 find such pearls in long CoT sequences。为了验证我们的动机, 我们迫使模型在思维过程的不同转换点从思考切换到直接生成答案。如果获得的答案是正确的, 则验证了这种“珍珠推理”的存在。如图1所示, 大约 75% 的样本包含这种珍珠 (提前退出能产生正确答案), 甚至有 36.7% 的样本仅需要不到原始推理路径的一半就能得出正确结论。因此, 如何找到珍珠推理是实现高效推理的一个有价值的研究课题。

当发现推理转换点时, 会触发动作 2) Trial Answer Inducing: 我们将其替换为 “final answer” 标记, 以诱导模型立即生成一个试验性答案, 该答案将用于 3) Confidence Evaluating。如果置信度足够高, 则设置模型停止进一步思考, 并根据已生成的思路得出结论。否则, 撤销“诱导试验性答案”的动作, 模型继续沿着原始路径进行推理。此外, 考虑到模型对答案诱导提示的潜在敏感性, 我们提出了 DEER-Pro (DEER 的 P 行且 Ro 健变体), 它在潜在的早期退出点执行多个并行的答案诱导, 并根据聚合结果校准置信度, 从而进一步确保 DEER 的鲁棒性。

我们的方法简单而有效, 并且可以无缝扩展到十一种不同架构和规模的推理模型, 在十个推理基准测试中取得了优异的结果, 包括数学任务 (例如 AIME 2024、AMC 2023 和 MATH-500)、科学任务 (例如 GPQA Diamond) 和编程任务 (例如 BigCodeBench)。具体来说, 我们的方法在集成到前沿推理模型中时, 可以将思维链序列的长度平均减少 19.1% 到 80.1%, 同时在不同推理基准测试中将准确率提高 0.3% 到 5.0%。我们的 DEER 为提升 LRM 的效率和准确性提供了一个即插即用的解决方案。

2 Motivations and Observations

在本节中, 我们分析 LRM 中的过度思考现象, 并研究静态早期退出对模型性能的影响。我们将“珍珠推理”定义为推理信息恰好足以准确解决问题的关键节点。我们在图1中的分析显示, 大约 75% 的样本包含此类珍珠 (即早期退出能产生正确答案)。此外, 我们识别出了一个样本子集, 对于这些样本, 正确答案只能通过早期退出获得 (如图1中的问题 11、19 和 26 所示)。图2(a) 中呈现的定量分析进一步表明, 在 MATH-500 和 GPQA 中, 分别有 60.8% 和 35.1% 的正确回答样本, 在仅完成 20% 的推理步骤后采用早期退出时, 仍能保持其准确性。这些实证结果证实了我们的假设: LRM 有潜力通过策略性的早期终止, 在计算效率和预测准确性上同时实现改进。

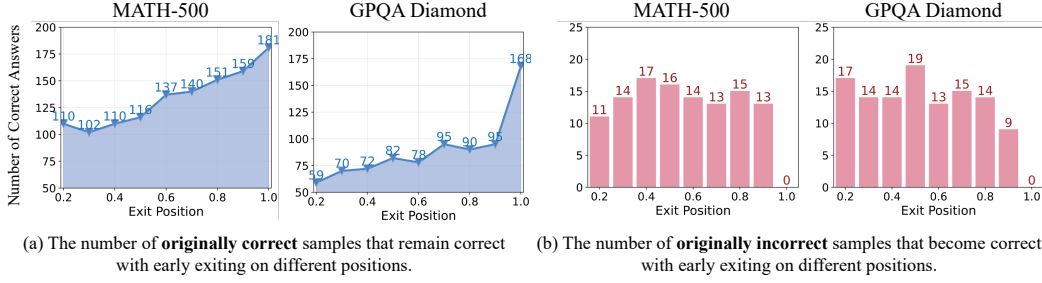


图 2: 定量试点实验结果。具体设置请参阅附录A。

图2(b) 表明，在不同位置退出能纠正不同比例的错误答案。对于 MATH 数据集，在 40% 的推理步骤处退出时，纠正率最高；而对于 GPQA 数据集，最佳纠正是在 50% 处退出时发生。每个问题的最佳早期退出点各不相同，并且与问题本身的内在难度密切相关。因此，直观上，依赖基于固定启发式的静态早期退出策略是次优的，这突显了设计动态早期退出机制的必要性。

3 Method

3.1 The Generation Pattern of Large Reasoning Models

与传统的大型语言模型（System 1）相比，大型推理模型（System 2）(Li et al., 2025b) 在推理阶段展现出独特的生成模式。（1）LRMs 使用分隔符将输出划分为两个过程：慢思考和结论。LRMs 在慢思考中进行系统且彻底的推理，最终总结思考过程并在结论中给出最终答案。（2）在慢思考过程中，LRMs 进行复杂的思考动作（thoughts），例如问题理解、方法探索和结果验证 (Luo et al., 2025b)。在每个推理动作（thought）内部，模型执行特定的程序性动作，而不同推理动作之间的转换通常由动作转换点（ATP）标记，例如“Wait”、“Alternatively”。

$$\text{System 1: [Prompt] + [Completion],} \quad (1)$$

$$\text{System 2: [Prompt] + \langle \text{think} \rangle + [\text{Slow Thinking}] + \langle / \text{think} \rangle + [\text{Conclusion}],} \quad (2)$$

$$[\text{Slow Thinking}] : [\text{Action Execution}] + (\text{ATP}) + [\text{Action Execution}] + (\text{ATP}) + \dots, \quad (3)$$

其中 $\langle \text{think} \rangle$ 和 $\langle / \text{think} \rangle$ 分别是思考开始和思考结束的分隔符。

在本节中，我们介绍动态推理早期退出（DEER）方法，以确定早期退出（珍珠推理路径）的最佳位置，从而缓解过度思考问题。

DEER 背后的核心思想是，模型对其试探性答案的置信度动态地表明 LRMs 生成最终答案所需的思考信息是否充足。我们观察到，当模型的推理过程不完整或有缺陷时，试探性答案的置信度往往显著较低。相反，当推理全面且逻辑合理时，模型会生成置信度更高的答案，如图17所示。这表明模型隐含地认识到何时发生 pearl reasoning，但在推理过程中缺乏明确的机制来利用这种意识进行早期终止。DEER 旨在通过将隐含意识转化为明确的早期退出决策来弥合这一差距。

如图3所示，DEER 涉及三种设计来决定是否提前退出：推理转换监控器、答案诱导器和置信度评估器。Reasoning transition monitor. 在 DEER 框架内，我们提出了两种可选的监控器设计策略：(i) linguistic marker-based，以及 (ii) entropy-based 监控。对于第一种策略，如第 3.1 节所述，LRMs 明确利用 ATPs 来标记不同思维之间的边界。这一特性使得 DEER 能够将 ATPs 识别为潜在的提前退出机会。在第二种策略中，DEER 使用 “\n\n” 作为分

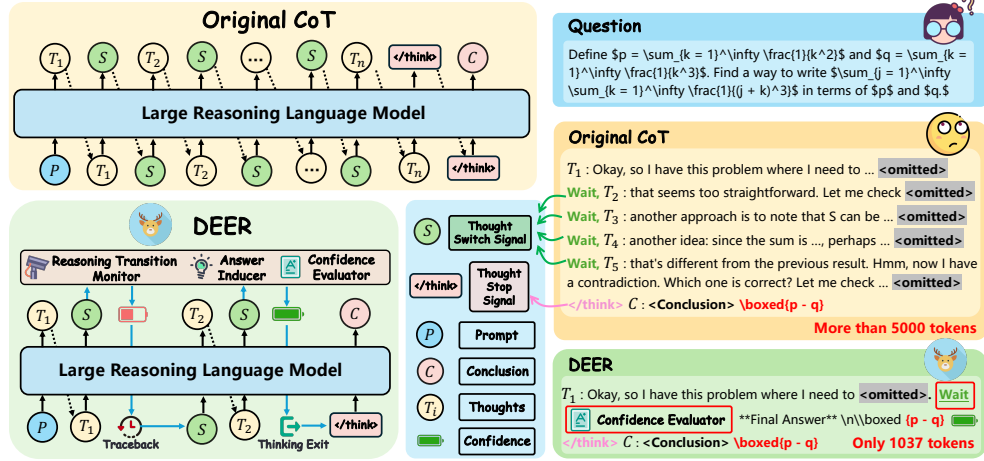


图 3: 动态推理早期退出 (DEER) 方法概述。

隔符来划分推理步骤。在每个推理步骤之后，DEER 计算初始标记的熵，记为 $H(p(\cdot|x_{<t}))$ 。低熵值表明模型正在进行程序性动作执行，其特点是推理模式稳定。相反，高熵值表明模型正在思考其后续的推理动作，多个潜在路径被同时激活。这些表现出高熵的位置被识别为提前退出的候选点。

我们在第 ?? 节中的后续实验表明，使用外部语言标记的 DEER 满足与第二种基于内部状态的方法相似的性质，同时实现了可比的性能。当应用于英语 LRMs 时，现有模型一致表现出生成此类语言标记的模式。遵循奥卡姆剃刀原则 (Rasmussen & Ghahramani, 2000)，我们建议采用第一种策略。对于非英语推理场景，备选的第二种策略也能准确捕捉提前退出点，这证明了 DEER 的通用性和鲁棒性。

Answer inducer. 当 LRM 在一个潜在的提前退出点暂停时，试验答案诱导器模块会提示模型根据目前已生成的推理内容生成一个中间答案。我们将答案分隔符 ($\boxed{\cdot}$) 纳入提示中，以便更精确地识别试验答案，如下所示： $\mathbf{A} = \text{LRM}(\mathbf{P}, \mathbf{T}, \mathbf{I})$

(4)

其中 $\mathbf{P}$ 表示输入提示， $\mathbf{T}$ 表示生成的思维， $\mathbf{I}$ 表示答案诱导器提示，而 $\mathbf{A} = [a_0, a_1, \dots, a_n]$ 是试验答案。**Confidence evaluator.** 置信度评估器计算诱导出的试验答案的置信度。它取每个词元的最高预测概率作为其置信度。对于多词元的试验答案，其整体置信度计算为所有词元得分的平均值，如下所示：

$$p(a_t) = \text{softmax}(\mathcal{M}(\mathbf{P}, \mathbf{T}, \mathbf{I}, \mathbf{a}_{<t})), \quad \mathcal{C} = \left(\prod_{i=1}^n \max_{a_t \in \mathcal{V}} p(a_t) \right)^{1/n} \quad (5)$$

其中 $\mathcal{M}$ 是 LRM 最后一层的语言模型头。 $\mathcal{C}$ 的计算采用几何平均数，这更好地契合了联合概率的乘法性质，并对低概率值表现出更高的敏感性，从而提供了更强的鲁棒性。

最后，将获得的置信度与经验阈值 λ 进行比较，以决定是否提前退出。如果 $\mathcal{C} > \lambda$ ，我们认为 LRM 当前生成的推理信息已足够，表明模型已达到 pearl reasoning。此时，DEER 停止进一步的推理动作，并继续给出结论。否则，模型将回退到上一个转换点以生成下一个思考。

DEER-PRo. 为了进一步提高珍珠推理识别的可靠性和准确性，我们引入了 DEER-PRo，它是 DEER 的一个 P 并行且 Ro 鲁棒的变体。通过在提前退出点使用不同的提示进行答案诱导，DEER-PRo 计算多个置信度值的平均值和平均绝对偏差 (MAD)，从而得出一个经过

校准的置信度分数 C_{cali} ，如下所示：

$$C_{cali} = C_{avg} - \alpha \cdot C_{MAD}, \quad C_{avg} = \frac{1}{N} \sum_{i=1}^N C_i, \quad C_{MAD} = \frac{1}{N} \sum_{i=1}^N |C_i - C_{avg}| \quad (6)$$

其中 $C_i = \left(\prod_{t=1}^n \max_{a_t \in \mathcal{V}} \text{softmax}(\mathcal{M}(\mathbf{P}, \mathbf{T}, \mathbf{I}_i, \mathbf{a}_{<t}^i)) \right)^{1/n}$ 表示使用特定答案诱导提示 $\mathbf{I}_i$ 获得的置信度分数， N 表示诱导尝试的次数， α 是波动惩罚强度系数。引入的保守偏差 C_{MAD} 有效地防止了由于提示中的正向噪声导致置信度分数被高估（即估计置信度超过真实置信度）而引起的错误提前退出。我们在附录 ?? 中证明， C_{cali} 有效地消除了模型对答案诱导提示的敏感性对提前退出准确性的影响，从而显著提高了 DEER 的鲁棒性。

直观上，DEER 中答案诱导器和置信度评估器的计算在推理过程中引入了额外的延迟，尤其是在代码生成任务中，试验答案仍然很长。这种开销削弱了通过大幅减少生成的 CoT 序列所实现的效率提升。为了解决这一挑战，我们将 DEER 与分支并行加速策略（图9）相结合，该策略通过两个关键机制缓解了这些效率限制：(1) 使用专门的因果注意力掩码将多个分支线性化为单个序列以进行并行生成，以及 (2) 通过基于置信度的剪枝进行动态 KV 缓存管理。该策略促进了试验答案评估与并发推理链生成之间的时间重叠，从而优化了整体推理效率。

4 Experiments

4.1 Experimental Setup

Benchmarks, Metrics and Implementations. 我们在 10 个基准测试上评估模型性能，包括 6 个数学推理基准：GSM8K (Cobbe et al., 2021)、MATH-500 (Hendrycks et al., 2021)、AMC 2023 (AI-MO, 2024)、AIME 2024、AIME 2025 (MAA Committees)、OlympiadBench (He et al., 2024)，一个科学推理基准：GPQA Diamond (Rein et al., 2023)，以及 3 个代码推理基准：HumanEval (Chen et al., 2021)、BigCodeBench (Zhuo et al., 2024) 和 LiveCodeBench (Jain et al., 2024)。在六个数学推理基准中，GSM8K、MATH-500 和 AMC 2023 通常被认为是相对简单的推理任务，而 AIME 2024、AIME 2025 和 OlympiadBench 则被视为更具挑战性。鉴于评估基准集广泛，我们在主要实验中选择性呈现最流行的几个（GSM8K、MATH-500、AMC 2023、AIME 2024 和 GPQA Diamond）。更多实验结果在附录 ?? 中提供。我们选择 Accuracy (Acc)、Token Number (Tok) 和 Compression Rate (CR) 作为评估指标。Acc 表示最终答案的准确率。Tok 表示每个样本的平均生成长度，用于评估成本。CR 定义为平均响应长度与原始模型平均响应长度的比值，数值越低表示压缩率越高。鉴于数据集 AMC 2023、AIME 2024 和 AIME 2025 中的样本数量有限，我们对每个实例进行 4 轮采样，并对所有指标的结果取平均值，以确保稳定性和可靠性。我们已使用 HuggingFace Transformers (Wolf et al., 2020) 和 vLLM 推理加速框架 (Kwon et al., 2023) 实现了 DEER。本文呈现的实验结果基于 vLLM 实现。我们将超参数 λ 设置为 0.95 ($\lambda = 0.95$)。对于基于熵的 DEER，遵循 (Wang et al., 2025b) 中提出的 80/20 原则，我们将熵值超过 0.672 的推理步骤终止位置指定为提前退出点。对于 DEER-Pro，我们设置 $N = 4$ 和 $\alpha = 1$ 。更多实验设置细节置于附录 B 中。

Backbone LRMs and Baselines. 我们在开源的 DeepSeek-R1-Distill-Qwen 系列模型（1.5B、7B、14B 和 32B）(DeepSeek-AI et al., 2025)、Qwen3 系列模型（1.7B、4B、8B、14B、32B）(Qwen et al., 2025)、QwQ-32B (Team, 2025) 以及 DeepSeek-R1 (Liu et al., 2025c) 上进行了实验。由于评估的模型数量众多，我们在主实验中选取 DeepSeek-R1-Distill-Qwen-7B、Qwen3-14B 和 QwQ-32B 作为代表性示例。更多实验结果在附录 B 中提供。我们将 DEER

表 1: 各类推理模型的实验结果。”Acc”表示准确率, ”Tok”表示词元数量, ”CR”表示压缩率。↑表示数值越高越好, 而 ↓表示数值越低越好。最佳结果以 bold 高亮显示。

Method	Math												Science			Overall	
	GSM8K			MATH-500			AMC23			AIME24			GPQA-D				
	Acc↑	Tok↓	CR↓	Acc↑	Tok↓	CR↓	Acc↑	Tok↓	CR↓	Acc↑	Tok↓	CR↓	Acc↑	Tok↓	CR↓	Acc↑	CR↓
DeepSeek-R1-Distill-Qwen-7B																	
Vanilla	89.6	1,484	100%	87.4	3,858	100%	78.8	6,792	100%	41.7	13,765	100%	23.7	10,247	100%	64.2	100%
TCC	88.0	892	60.1%	89.2	3,864	100.2%	82.5	6,491	95.6%	48.4	10,603	77.0%	27.3	8,442	82.4%	67.1	83.0%
CoD	84.7	298	20.1%	83.2	1,987	51.5%	77.5	4,440	65.4%	40.0	10,519	76.4%	37.9	6,431	62.8%	64.7	55.3%
NoThinking	87.1	284	19.1%	80.6	834	21.6%	65.0	1,911	28.1%	26.7	4,427	32.2%	29.8	724	7.1%	57.8	21.6%
Dynasor-CoT	89.6	1,285	86.6%	89.0	2,971	77.0%	85.0	5,980	88.0%	46.7	12,695	92.2%	30.5	7,639	74.5%	68.2	83.7%
SEAL	88.4	811	54.6%	89.4	2,661	69.0%	—	—	—	—	—	—	—	—	—	—	—
DEER	90.6	917	61.8%	89.8	2,143	55.5%	85.0	4,451	65.5%	49.2	9,839	71.5%	31.3	5,469	53.4%	69.2	61.5%
DEER-PRo	91.0	989	66.7%	90.2	2,391	62.0%	87.5	4,877	71.8%	49.2	10,046	73.0%	30.6	5,682	55.5%	69.7	65.8%
Qwen3-14B																	
Vanilla	95.1	2,047	100%	93.8	4,508	100%	95.0	7,139	100%	70.0	10,859	100%	60.1	7,339	100%	82.8	100%
TCC	95.7	1,241	60.6%	94.6	4,484	99.5%	95.0	7,261	101.7%	70.8	11,573	106.6%	60.1	7,138	97.3%	83.3	93.1%
CoD	85.7	648	31.7%	75.2	2,359	52.3%	72.5	4,122	57.7%	60.0	10,768	99.2%	51.0	1,177	16.0%	68.9	51.4%
NoThinking	94.8	286	14.0%	85.0	1,228	27.2%	77.5	2,133	29.9%	26.7	7,337	67.6%	50.5	2,320	31.6%	66.9	34.1%
Dynasor-CoT	95.6	1,483	72.4%	93.8	4,063	90.1%	95.6	6,582	92.2%	73.3	10,369	95.5%	59.6	5,968	81.3%	83.6	86.3%
DEER	95.3	840	41.0%	94.0	3,074	68.2%	95.0	4,763	66.7%	76.7	7,619	70.2%	57.6	2,898	39.5%	83.7	57.1%
DEER-PRo	95.3	926	45.2%	94.4	3,260	72.3%	95.6	4,905	68.7%	75.0	8,135	74.9%	61.2	4,062	55.4%	84.3	63.3%
QwQ-32B																	
Vanilla	96.7	1,427	100%	93.8	4,508	100%	92.5	6,792	100%	66.7	10,821	100%	63.1	7,320	100%	82.6	100%
TCC	95.8	1,348	94.5%	94.4	4,315	95.7%	90.0	6,818	100.4%	60.0	11,263	104.1%	61.6	7,593	103.7%	80.4	99.7%
CoD	96.0	627	43.9%	94.0	3,630	80.5%	92.5	5,943	87.5%	60.0	10,731	99.2%	62.6	6,039	82.5%	81.0	78.7%
NoThinking	96.2	1,113	78.0%	94.8	3,930	87.2%	87.5	6,908	101.7%	66.7	10,859	100.4%	63.6	7,668	104.8%	81.8	94.4%
Dynasor-CoT	95.2	1,095	76.7%	94.2	4,176	92.6%	93.8	6,544	96.3%	63.3	11,156	103.1%	64.1	7,024	96.0%	82.1	93.0%
DEER	96.3	977	68.5%	94.6	3,316	73.6%	95.0	5,782	85.1%	70.0	10,097	93.3%	64.1	6,163	84.2%	84.0	80.9%
DEER-PRo	96.2	1032	72.3%	94.8	3,650	80.9%	95.0	5,811	85.6%	70.0	10,264	94.9%	64.7	6,201	84.7%	84.1	83.7%

与现有的基于提示和基于输出的高效推理方法进行比较, 包括 Vanilla、TCC (Muennighoff et al., 2025)、CoD (Xu et al., 2025c)、NoThinking (Ma et al., 2025a)、Dynasor-CoT (Fu et al., 2025) 以及 SEAL (Chen et al., 2025a)。Vanilla 对 LRM 进行直接评估, 不做任何干预。令牌条件控制 (TCC) 在系统提示中指定固定的令牌数量以强制执行令牌预算; 在我们的实验中, 我们根据 DEER 生成的实际令牌长度来设置此限制。链式草稿 (CoD) 通过限制每个推理步骤中使用的词语数量来减少冗长, 仅关注推进所需的必要计算或转换。NoThinking 提示模型跳过推理阶段, 直接生成最终答案。Dynasor-CoT 定期提示模型在固定的令牌间隔处生成中间答案, 并在三个连续答案一致时触发提前退出。SEAL 训练一个转向向量来校准 CoT 过程, 引导模型进行更可靠的推理。

4.2 Main Results

Overall Performance. 由于篇幅限制, 表1展示了五个广泛采用的推理基准, 在三个最先进的推理模型上进行了评估, 这些模型专门涵盖了三种模型规模, 全面证明了 DEER 的优越性能。我们还在附录中提供了涵盖 10 个数据集、11 个参数规模从 1.5B 到 671B 的模型的更多结果。可以发现, DEER 在各种推理模型和任务中表现出强大的适应性, 与原始模型相比, 在将序列长度减少 19.1% 至 42.9% 的同时, 实现了 0.9 到 4.8 个百分点的准确率提升。与 DEER 相比, DEER-Pro 仅以 2.8% 至 6.2% 的生成长度边际增加, 就实现了更高的准确率。我们在额外的较小规模模型上进行了 DEER 与 DEER-Pro 的对比实验。表2中的实验结果表明, DEER-Pro 实现了更显著的准确率提升。这表明 DEER-Pro 有效解决了较小模型中的提示敏感性问题, 展现了其卓越的鲁棒性。

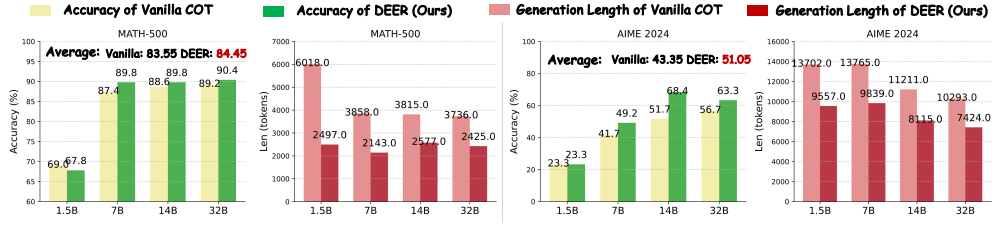


图 4: DEER 与 Vanilla CoT 在不同规模的 DeepSeek-R1-Distill-Qwen-Series 模型上, 在 MATH-500 和 AIME 2024 上的实验结果对比。

Comparison with Efficient Reasoning SoTAs. 表 1 展示了 DEER 与近期高效推理方法之间的比较。可以观察到, DEER 始终优于所有基线方法, 而基线方法要么难以在不同任务和基础模型之间泛化, 要么必须在准确性和效率之间进行权衡。具体来说, 虽然 TCC Muennighoff et al. (2025) 通过将令牌预算纳入提示, 在 GSM8K 等较简单的任务上实现了合理的效率-准确性权衡, 但在复杂问题 (如 AIME24) 上却失败了, 因为模型忽略了提示的长度限制, 生成了比普通 CoT 更长的回答。至于 NoThinking 和 CoD, 虽然实现了显著的长度缩减, 但它们严重损害了模型固有的推理能力。相比之下, Dynasor-CoT 保持了推理质量, 但由于其保守的提前退出条件导致终止过晚, 导致长度缩减微乎其微。值得注意的是, 几乎所有基线方法在 QwQ-32B 上都完全失败, 原因是其思维结束分隔符 `</think>` 会偶发性失效, 模型在其后继续生成推理步骤, 并经常产生重复的 `</think>` 令牌 (如附录图 18 所示)。值得注意的是, 尽管存在这些挑战, DEER 在 QwQ-32B 上仍然实现了 19.1% 的长度缩减, 进一步证明了其鲁棒性。

Performance on Programming Tasks. 表 3 报告了 DEER 在三个编程任务上的评估结果, 完成了我们对推理模型三个主要领域 (数学、科学和编程) 的全面覆盖。它展示了 DEER 在不同编程任务和模型规模上的一致有效性, 与数学和科学任务相比, 实现了更小的压缩率 (平均 19.9% 对比 61.5%)。这种增强的压缩可能源于代码生成的固有特性, 其中每个推理步骤通常会产生包含大量冗余令牌的冗长代码段。

Performance Trends across Model Sizes and Reasoning Difficulty. 图 4、10 展示了在 MATH-500 和 AIME 2024 数据集上的评估结果, 以检验 DEER 在不同模型规模上的性能增益。可以看出, DEER 在所有模型规模上都能持续提高准确性, 同时减少令牌消耗。一个关键的观察是, 较小的模型 (例如 1.5B) 倾向于生成明显更长的推理序列, 并表现出更严重的过度思考现象。这源于它们在 CoT 生成过程中发现正确推理步骤的有限能力。因此, 我们的方法对这些较小的模型实现了更大的长度缩减。图 4 使用 MATH-500 (简单推理) 和 AIME 2024 (挑战性推理) 数据集作为代表性基准。结果表明了 DEER 的双重能力: 它在更简单的问题上实现了更优越的压缩率, 同时在复杂任务上带来了更显著的准确性提升。这恰好解决了推理系统中的两个关键需求: 简单场景下的效率需求和挑战性场景下日益增长的准确性需求。

Performance Trends across Token Budgets. 图 5 评估了 DEER 在不同令牌预算 (由不同的最大长度设置控制) 下的性能。在子图 (a) 和 (f) 中, x 轴表示模型生成的思维链序列的实际长度, 而 y 轴表示模型准确率。准确率与效率之间的最佳平衡由靠近左上角的曲线所展示。蓝色阴影区域定量地表示了 DEER 的性能增益: 垂直高度对应准确率提升, 水平宽度对应令牌压缩收益。可以看出, DEER 始终优于原始方法, 因为所有点都位于原始方法对应点的左上方。如右侧四列子图所示, 我们观察到随着令牌预算增加, 原始模型生成长度更长、准确率更高的序列, 这证实了测试时缩放效应。值得注意的是, DEER 展示了一种自适应的权衡: 在受限的令牌预算下, 它在准确率上获得更大增益, 但在长度压缩方面的收益减

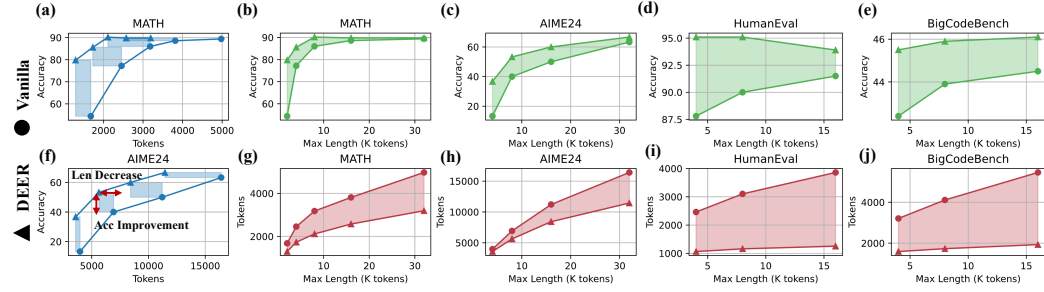


图 5: 基于 DeepSeek-R1-Distill-Qwen-14B 模型, DEER 与基线方法在四个数据集上、不同令牌预算设置下的性能比较。

少。相反, 在更大的令牌预算下则观察到相反的趋势。这表明我们的方法可以动态调整令牌预算, 以满足不同场景下对准确率与效率的不同要求。

4.3 Ablation Study

Impact of Reasoning Transition Monitor Choices. 在主要实验中, 我们采用 “Wait” 作为早期退出的监控信号, 记作 DEER(W)。这种基于简单语言标记的方法取得了有希望的结果。为了比较不同早期退出信号对 DEER 性能的影响, 我们进行了额外的实验, 使用 “Alternatively” 作为信号, 以及基于熵的监控来进行早期退出检测。相应的结果展示在表 5 和 6 中。表 5 收集了通过将原始 CoT 在潜在退出点处划分所得推理块的数量和平均长度的统计数据。块的数量表明 DEER(Ent) 提供了最多的早期退出机会, 而 DEER(A) 提供的最少, 这与平均生成长度呈负相关。表 6 中在额外数据集和模型上的结果表明, 基于熵的监控和基于语言标记的监控都表现出可比且优越的性能, 显著优于基线。在大规模实际部署中, 鉴于其实现的简单性和高效性, 我们提倡使用基于语言标记的方法。附录 G 对这两种监控策略进行了进一步的探讨。

Robustness of threshold λ . 图 6 展示了 DEER 在 MATH-500 数据集上使用不同阈值 λ 的性能。结果表明, 当阈值设置过低时, 推理长度的微小额外减少会导致准确率显著下降, 这反映了对过度思考的过度校正。相反, 当阈值设置过高时, 模型退出推理过晚, 导致推理时间延长且准确率下降。此外, 可以看出我们的方法在 0.9-0.97 范围内对 λ 具有鲁棒性, 无需进行超参数调优。表 4 展示了我们使用 Qwen3 在额外数据集上对阈值 λ 的鲁棒性研究。实验结果表明, Qwen3 对 λ 表现出卓越的鲁棒性, 在 0.8-0.97 范围内始终保持强劲性能。此外, 附录中在 11 个模型和 10 个数据集上进行的实验结果, 均统一采用 0.95 作为阈值。持续强劲的结果进一步证明了 DEER 的泛化能力和鲁棒性。附录第 H 节揭示了 DEER 鲁棒性的根本来源在于置信度极化现象。

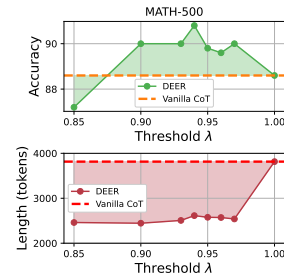


图 6: λ 的影响。

4.4 Discussion

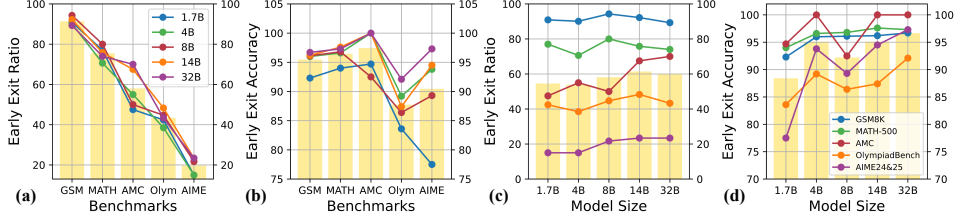


图 8: DEER 的提前退出率及提前退出样本的准确率。图 (a) 和 (b) 共享一个图例，图 (c) 和 (d) 亦然。每个条形的高度反映了平均值。

Efficiency Improvement. 为了准确验证 DEER 及其分支并行加速变体在端到端推理效率上带来的增益，我们基于 huggingface transformers (Wolf et al., 2020) 在 MATH-500 和 AMC 2023 数据集上测量了平均延迟。如图 7 (a) 所示，原始 DEER 将延迟降低了 27.9% 至 40.1%，而提出的分支并行解码变体将延迟降低了 36.3% 至 58.6%。这表明分支并行 DEER 通过有效减少

试验答案生成和置信度评估的延迟，实现了进一步的加速提升。此外，我们针对 MATH-500 上的每个样本，将延迟加速比与长度节省量进行了映射。图 7 (b) 表明，延迟加速比与长度节省量之间的比率呈现出超线性趋势，这强化了 DEER 在提升推理速度方面的重要性。在附录的章节 F 中，我们从理论上证明了 DEER 的效率，并解释了超线性加速的根本原因。

Exploring the Effectiveness of DEER’s Early-Exit Mechanism. 图 8 展示了不同规模的 Qwen3 系列模型的早退率以及早退样本的准确率。如图 8(a) 所示，DEER 的早退率随着任务难度的增加而降低，这解释了其在复杂任务上相比简单任务压缩性能相对较低的原因。图 8(b) 显示，尽管随着任务难度增加，早退准确率有所下降，但其始终保持在较高水平——范围从 88% 到 98%。在一项同期研究中，Zhang et al. (2025a) 训练了一个探针来决定是否早退。然而，他们的探针在 MATH-500 上的准确率仅为 80% 左右，显著低于 DEER 的 95%。这表明模型本身具备评估答案正确性的能力，而 DEER 有效地利用了这种能力。如图 8(c) 和 (d) 所示，尽管在不同难度的任务间存在微小差异，但总体上存在随着模型规模增大，早退率更高、准确率提升的趋势。这一观察结果意味着 DEER 的性能与模型容量之间存在正相关关系：更大的模型能产生更准确的置信度估计，从而做出更好的早退决策。我们进一步研究了 DEER 获得准确率提升的原因。图 11 表明，DEER 通过早退纠正的答案（绿色条）多于其错误改变的答案（红色条）。这表明 DEER 不仅通过在能正确回答的问题上早退来节省计算成本，还能纠正有问题的思维过程。

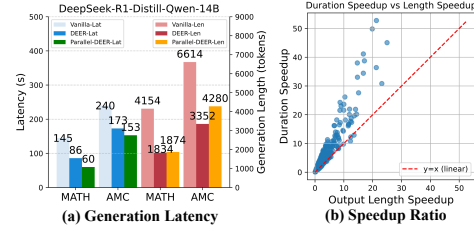


图 7: 效率提升。

Case Study. 图 13 显示，DEER 和原始的 CoT 都在第一个推理步骤中得出了正确答案，如绿色框所示。区别在于，DEER 在评估试验答案的置信度足够高后提前退出，从而产生了正确结果。相比之下，原始的 CoT 则继续进行下一个推理动作。在双重检查并切换推理方法后，由于两种方法得出的答案不一致，模型陷入了无休止的验证循环，最终未能提供最终答案。此外，图 17 表明，

LRMs 隐含地知道何时提前退出，而我们的方法简单有效，能够实现模型自身的这种潜力。详情请参阅附录 I。

5 Related Work

遵循 (Sui et al., 2025; Wang et al., 2025a) 中建立的高效推理分类法，我们将相关工作分为三类：post-training based 方法使用带有可变长度思维链数据的 SFT (Yu et al., 2024; Kang et al., 2025; Xia et al., 2025; Ma et al., 2025b; Munkhbat et al., 2025; Liu et al., 2024; Han et al., 2024) 或在强化学习中引入长度奖励 (Team et al., 2025b; Luo et al., 2025a; Aggarwal & Welleck, 2025; Arora & Zanette, 2025; Yeo et al., 2025; Shen et al., 2025b; Qu et al., 2025; Cui et al., 2025)，使模型能够自适应地生成不同长度的思维链，这超出了我们无需训练的范围。Prompt-based 方法 (Han et al., 2024; Xu et al., 2025b; Lee et al., 2025; Renze & Guven, 2024; Chen et al., 2024) 使用不同的提示来强制推理模型生成简洁的思维链，减少不必要的推理步骤。Output-based 方法旨在加速模型解码阶段的推理生成，DEER 属于此类。然而，大多数先前工作 (Xie et al., 2023; Liao et al., 2025; Li et al., 2024; Manvi et al., 2024; Aggarwal et al., 2023) 侧重于优化最佳 N 采样，这与我们的研究无关。相反，我们选择了三个近期的并行工作 Nothinking (Ma et al., 2025a)、Dynasor-CoT (Fu et al., 2025) 和 SEAL (Chen et al., 2025a) 作为比较基线。更多相关工作可见附录 J。

6 Conclusion

本文验证了思维链生成中提前退出动机背后的原理，并据此提出了一种无需训练的、动态的提前退出算法，该算法使推理模型在思考量恰好足够时退出后续思考。我们的方法在不同模型规模的推理模型上进行了全面评估，并在十个经典推理基准上以更少的标记数展示了优越性能，为测试时扩展中常见的准确性与效率权衡提供了一个双赢的解决方案。

7 Ethics statement

本工作遵循 ICLR 道德准则。我们确认我们的研究在所有阶段都以诚信、诚实和尊重道德原则的方式进行。

- 本文中呈现的所有发现均被准确、诚实地报告。我们未捏造、伪造或歪曲任何数据或结果。我们的方法和实验过程描述透明，以确保可复现性。
- 本研究中使用的所有数据集均根据其许可证和使用条款获取和使用。对于任何涉及个人信息的数据，我们确保遵守隐私法规，并在必要时获得了适当的伦理批准。
- 对本工作的所有贡献均已得到适当承认。我们恰当地引用了所有影响我们研究的来源和先前工作。所有合著者均对本工作做出了实质性贡献，并同意提交。

- 我们已仔细考虑了本工作更广泛的影响。虽然我们的研究旨在积极推动该领域发展，但我们承认潜在的军民两用关切，并鼓励在现实世界应用中负责任地部署我们的方法。

8 Reproducibility statement

为确保我们结果的可复现性，我们提供了实验设置、超参数和实现细节的详细描述。在可能的情况下，我们提供了代码和补充材料，以便于验证和未来的研究。

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A Pilot Experiment Setup

我们选择 AIME2024 (MAA Committees) 作为探索性实验的测试集进行定性分析，并进一步通过在 MATH-500 (Hendrycks et al., 2021)、GPQA-Diamond (Rein et al., 2023) 上的实验进行定量分析。所有实验均在 DeepSeek-R1-Distill-Qwen-14B (DeepSeek-AI et al., 2025) 上进行。在我们的实验中，我们首先让 LRM 在测试集上执行完整的推理（包括慢思考内容和结论内容）。然后，我们保留思考内容，并根据动作转换点将其划分为思考块。保留思考块数量超过五个的样本。对于这些样本，我们保留了不同比例（20%-90%）的思考块，并在每个截断的推理序列后附加一个思考结束标记分隔符，以强制终止慢思考过程。模型随后基于部分推理内容生成其最终结论。对于在不同思考内容下获得的结论，我们评估了其正确性，并在图 1 中展示了每个样本的结果。此外，我们在图 2 中研究了在三个数据集上，原本正确的样本在提前退出后仍保持正确的数量，以及原本不正确的样本在提前退出后变为正确的数量。

(7)

A.1 Noise Independence of the MAD-Calibrated Strategy

让我们将真实置信度 μ 定义为模型在给定当前推理内容下得出正确答案的实际概率，对应于珍珠推理在此处存在的真实概率。提前退出决策阈值记为 λ ，当 $\mu > \lambda$ 时模型执行提前退出。由于答案诱导提示可能引入 ε ，模型的输出置信度围绕真实置信度波动，产生一个观测置信度 $C_i = \mu + \varepsilon_i$ 。在实际测试环境中，我们比较 C_i 与 λ 来决定是否执行提前退出。不失一般性，我们假设噪声项 ε_i 是独立同分布的，服从均值为 0、标准差为 σ 的高斯分布，即 $\varepsilon_i \sim N(0, \sigma^2)$ 。这里， σ 表示模型对提示措辞的敏感性。更大的 σ 意味着更高的模型敏感性，从而导致更大的置信度波动。

接下来，我们通过比较 DEER-PRO (C_{cali})、DEER (C_i) 和一种平均方法 (C_{avg}) 在关键风险场景下的决策错误率，来展示 DEER-PRO 对抗噪声干扰的有效性。假设 C_{cali} 和 C_{avg} 各自并行进行 N 次答案诱导，且 N 与方程 (5) 中的相同。鉴于在推理场景中保持准确性优先于提前退出的加速收益，我们将风险场景指定为真实置信度 $\mu < \lambda$ 的情况。然后我们计算由于噪声干扰而观测到 $C > \lambda$ 的假阳性概率。

A.1.1 Probability of Error for a Single Prompt:

$$P_{FP}(\text{Single}) = P(C_i > \lambda) = P(\mu + \varepsilon_i > \lambda) = P(\varepsilon_i > \lambda - \mu) \quad (8)$$

由于 $\varepsilon_i \sim N(0, \sigma^2)$ ，我们可以将其标准化为：

$$P_{FP}(\text{Single}) = P\left(\frac{\varepsilon_i}{\sigma} > \frac{\lambda - \mu}{\sigma}\right) \quad (9)$$

令 $Z = \frac{\varepsilon_i}{\sigma}$ ，则 $Z \sim N(0, 1)$ ，于是：

$$P_{FP}(\text{Single}) = P\left(Z > \frac{\lambda - \mu}{\sigma}\right) = 1 - \Phi\left(\frac{\lambda - \mu}{\sigma}\right) \quad (10)$$

其中 Φ 是标准正态分布的累积分布函数。

A.1.2 Probability of Error for Averaged Confidence:

对于平均置信度 $\mathcal{C}_{avg}$ ，噪声项仍然是高斯的， $\varepsilon_{avg} \sim N(0, \sigma^2/N)$ 。平均置信度的错误概率为：

$$P_{FP}(\text{Avg}) = P(\mathcal{C}_{avg} > \lambda) = 1 - \Phi\left(\sqrt{N} \frac{(\lambda - \mu)}{\sigma}\right) \quad (11)$$

由于对于 $N > 1$ 有 $\frac{(\lambda - \mu)\sqrt{N}}{\sigma} > \frac{\lambda - \mu}{\sigma}$ ，因此可得 $P_{FP}(\text{Avg}) < P_{FP}(\text{Single})$ 。这表明，简单地平均多个观测到的置信度值可以减轻噪声干扰。然而，置信度平均未能解决根本问题，因为错误率 P_{FP} 仍然依赖于噪声标准差 σ ，并随着 σ 增大而单调增加。对于具有显著固有噪声（大的 σ ）的模型， Φ 函数内部的参数趋近于零，驱使 P_{FP} 趋向于 0.5。这表明，无论决策阈值 λ 如何，高噪声模型都会退化为随机猜测。因此，传统方法的可靠性严重受限于模型固有的噪声水平 σ ，这是一个我们无法控制的因素。

A.1.3 Probability

of Error for MAD-Calibrated Confidence (DEER-PRo):

$$P_{FP}(\text{calibration}) = P(\mathcal{C}_{cali} > \lambda) = P(\mathcal{C}_{avg} - \alpha \cdot \mathcal{C}_{MAD} > \lambda) \quad (12)$$

代入 $\mathcal{C}_{avg} = \mu + \varepsilon_{avg}$ ，我们得到：

$$P_{FP}(\text{calibration}) = P(\mu + \varepsilon_{avg} - \alpha \cdot \mathcal{C}_{MAD} > \lambda) \quad (13)$$

重新排列方程中的项，得到：

$$P_{FP}(\text{calibration}) = P(\varepsilon_{avg} > \alpha \cdot \mathcal{C}_{MAD} + \lambda - \mu) \quad (14)$$

接下来，我们将讨论 DEER-PRo 在两种不同场景下的鲁棒性。

Scenario 1: Approximate estimation of $\mathcal{C}_{MAD}$

在我们的假设下，其中 $\varepsilon_i \sim N(0, \sigma^2)$ ，我们有 $\varepsilon_{avg} \sim N(0, \sigma^2/N)$ 和 $E[\mathcal{C}_{MAD}] \approx 0.8\sigma$ （我们将在第 A.2 节提供证明）。对于大的 N ，大数定律允许我们将 $\mathcal{C}_{MAD}$ 近似为 0.8σ 。基于此假设，我们有：

$$P_{FP}(\text{calibration}) = P(\mathcal{C}_{avg} > \lambda + 0.8\sigma\alpha) \quad (15)$$

上述方程揭示了 DEER-PRo 的根本区别在于采用了随噪声缩放的自适应阈值：

$$\lambda_{\text{effective}} = \lambda + 0.8\alpha\sigma \quad (16)$$

我们继续通过将 $P_{FP}(\text{calibration})$ 转换为标准正态分布的累积分布函数 (CDF) Φ 来重新表述方程。对于方程 (14)，我们代入 $\mathcal{C}_{MAD} = 0.8\sigma$ 并得到：

$$P_{FP}(\text{calibration}) = P(\varepsilon_{avg} > 0.8\sigma\alpha + \lambda - \mu) \quad (17)$$

由于 $\varepsilon_{avg} \sim N(0, \sigma^2/N)$ ：

$$P_{FP}(\text{calibration}) = 1 - \Phi\left(\frac{(\lambda - \mu + 0.8\alpha\sigma)\sqrt{N}}{\sigma}\right) \quad (18)$$

简化：

$$P_{FP}(\text{calibration}) = 1 - \Phi\left(\sqrt{N} \left(\frac{\lambda - \mu}{\sigma} + 0.8\alpha\right)\right) \quad (19)$$

当噪声最小时：

$$\lambda_{\text{effective}} \rightarrow \lambda \quad (20)$$

校准方法的行为类似于标准阈值法，保持高效率。

当噪声占主导时， $\frac{\lambda - \mu}{\sigma} \rightarrow 0$ 。误报率变为：

$$P_{\text{FP}} = 1 - \Phi(0.8\alpha\sqrt{N}) \quad (21)$$

其独立于 σ 。这表明 DEER-PRo 有效地防止了提前退出退化为随机猜测 ($P_{\text{FP}} \rightarrow 0.5$)。

Scenario 2: Exact computation of $\mathcal{C}_{\text{MAD}}$ 无需近似。

从 $\mathcal{C}_{\text{MAD}} = \frac{1}{N} \sum_{i=1}^N |\mathcal{C}_i - \mathcal{C}_{\text{avg}}|$ ，我们知道 $\mathcal{C}_{\text{MAD}}$ 为正，因此将方程两边除以该项，我们得到：

$$P(\varepsilon_{\text{avg}}/\mathcal{C}_{\text{MAD}} > \alpha + (\lambda - \mu)/\mathcal{C}_{\text{MAD}}) \quad (22)$$

由于 $\mu < \lambda$ ，则 $\alpha + (\lambda - \mu)/\mathcal{C}_{\text{MAD}} > \alpha$ ，因此我们可以得到：

$$P(\varepsilon_{\text{avg}}/\mathcal{C}_{\text{MAD}} > \alpha + (\lambda - \mu)/\mathcal{C}_{\text{MAD}}) < P(\varepsilon_{\text{avg}}/\mathcal{C}_{\text{MAD}} > \alpha) \quad (23)$$

因此， $P(\varepsilon_{\text{avg}}/\mathcal{C}_{\text{MAD}} > \alpha)$ 是 $P_{\text{FP}}(\text{calibration})$ 的一个上界。对于比率 $\varepsilon_{\text{avg}}/\mathcal{C}_{\text{MAD}}$ ， ε_{avg} 代表噪声的信号，而 $\mathcal{C}_{\text{MAD}}$ 代表噪声的内部无序度。因此，我们可以将 $\varepsilon_{\text{avg}}/\mathcal{C}_{\text{MAD}}$ 定义为信噪比 (SNR)。接下来，让我们分析信噪比 (SNR) 的性质。在我们的假设下，当 $\varepsilon_i \sim N(0, \sigma^2)$ 时，我们有 $\varepsilon_{\text{avg}} \sim N(0, \sigma^2/N)$ 和 $E[\mathcal{C}_{\text{MAD}}] \approx 0.8\sigma$ 。由于 ε_{avg} 和 $\mathcal{C}_{\text{MAD}}$ 均与 σ 成比例，我们可以将 SNR 写为：

$$\text{SNR} = (\sigma * Z_{\text{avg}})/(\sigma * Z_{\text{MAD}}) = Z_{\text{avg}}/Z_{\text{MAD}} \quad (24)$$

其中 $Z_{\text{avg}} \sim N(0, 1/N)$ 是一个标准化的噪声均值，而 Z_{mad} 是一个与 MAD/σ 相关的随机变量，其分布不依赖于 σ 。因此，SNR 的概率分布独立于模型噪声标准差 σ 。

$$P_{\text{FP}}(\text{calibration}) < P(Z_{\text{avg}}/Z_{\text{MAD}} > \alpha) \quad (25)$$

因此， $P_{\text{FP}}(\text{calibration})$ 仅受提示数量 N 和信噪比阈值 α 的影响，其中更大的 N 和 α 值会导致更低的错误率。

通过从绝对阈值测试转变为自归一化的 SNR 测试，我们的 MAD 策略有效地将决策过程与模型固有的、不可控的噪声水平 σ 解耦。与传统方法在高噪声条件下失效不同，我们的方法提供了与模型噪声水平无关的一致、稳健的性能。

$$E[|z_i|] = \sqrt{\frac{2}{\pi}} \approx 0.798, \quad \text{Var}(|z_i|) = 1 - \frac{2}{\pi} \approx 0.363 \quad (26)$$

$$\text{Var}(Z_{\text{mad}}) \approx \frac{0.363}{N} \quad (27)$$

$$\text{Var}(\text{SNR}) \approx \left(\frac{\partial f}{\partial X} \right)_E^2 \text{Var}(X) + \left(\frac{\partial f}{\partial Y} \right)_E^2 \text{Var}(Y) \quad (28)$$

$$\left. \frac{\partial f}{\partial X} \right|_E = \frac{1}{0.798} \approx 1.25, \quad \left. \frac{\partial f}{\partial Y} \right|_E = 0, \quad \text{Var}(\text{SNR}) \approx (1.25)^2 \cdot \frac{1}{N} = \frac{1.5625}{N} \quad (29)$$

$$P(\text{SNR} > 1) \approx P\left(Z > \frac{\sqrt{N}}{1.25}\right) \quad (30)$$

在本节中，我们基于事件空间正式证明，MAD 策略的误报概率 $P_{\text{FP}}(\text{MAD})$ ($P_{\text{FP}}(\text{calibration})$) 显著优于简单平均策略 $P_{\text{FP}}(\text{Avg})$ 的误报概率，因此也优于 $P_{\text{FP}}(\text{Single})$ 。

Theorem. MAD 校准策略的误报概率满足 $P_{\text{FP}}(\text{MAD}) \leq \rho \cdot P_{\text{FP}}(\text{Avg})$ ，其中 $\rho = O(\exp(-\Theta(N)))$ 是关于提示数量 N 的指数衰减因子。

Proof. 误报事件定义如下：

$$E_{\text{Avg}} = \{\varepsilon : \varepsilon_{\text{avg}} > c\} \quad (31)$$

$$E_{\text{MAD}} = \{\varepsilon : \varepsilon_{\text{avg}} - \alpha \cdot \text{MAD} > c\} \quad (32)$$

其中 $c = \lambda - \mu > 0$ 是阈值间隙， $\text{MAD} = \frac{1}{N} \sum_{i=1}^N |\varepsilon_i - \varepsilon_{\text{avg}}|$ 是平均绝对偏差， $\varepsilon_{\text{avg}} = \frac{1}{N} \sum_{i=1}^N \varepsilon_i$ 是平均噪声，并且存在 N 个独立的噪声项 $\varepsilon_i \sim N(0, \sigma^2)$ ，其中 $i = 1, \dots, N$ 。

由于根据定义有 $\alpha > 0$ 和 $\text{MAD} \geq 0$ ，我们得到：

$$\varepsilon_{\text{avg}} - \alpha \cdot \text{MAD} \leq \varepsilon_{\text{avg}} \quad (33)$$

因此，如果 $\varepsilon_{\text{avg}} - \alpha \cdot \text{MAD} > c$ ，则必然有 $\varepsilon_{\text{avg}} > c$ 。这确立了：

$$E_{\text{MAD}} \subseteq E_{\text{Avg}} \quad (34)$$

因此：

$$P_{\text{FP}}(\text{MAD}) = P(E_{\text{MAD}}) \leq P(E_{\text{Avg}}) = P_{\text{FP}}(\text{Avg}) \quad (35)$$

为了量化超越这一基本不等式的改进程度，我们使用条件概率分解该概率：

$$P_{\text{FP}}(\text{MAD}) = P(E_{\text{MAD}} \cap E_{\text{Avg}}) = P(E_{\text{MAD}} | E_{\text{Avg}}) \cdot P(E_{\text{Avg}}) \quad (36)$$

由于 $E_{\text{MAD}} \subseteq E_{\text{Avg}}$ ，我们有：

$$P(E_{\text{MAD}} | E_{\text{Avg}}) = P(\varepsilon_{\text{avg}} - \alpha \cdot \text{MAD} > c | \varepsilon_{\text{avg}} > c) \quad (37)$$

这可以重写为：

$$P(E_{\text{MAD}} | E_{\text{Avg}}) = P\left(\text{MAD} < \frac{\varepsilon_{\text{avg}} - c}{\alpha} \middle| \varepsilon_{\text{avg}} > c\right) \quad (38)$$

定义改进因子：

$$\rho = P\left(\text{MAD} < \frac{\varepsilon_{\text{avg}} - c}{\alpha} \middle| \varepsilon_{\text{avg}} > c\right) \quad (39)$$

则有：

$$P_{\text{FP}}(\text{MAD}) = \rho \cdot P_{\text{FP}}(\text{Avg}) \quad (40)$$

为了评估 ρ ，我们分析满足 $\varepsilon_{\text{avg}} > c$ 的噪声向量结构。主要有两种模式：

Pattern A (Coherent Pattern): 所有噪声项都接近 c 。形式化地，对于某个小的 $\delta > 0$ ：

$$\text{Pattern A} = \{\varepsilon : |\varepsilon_i - c| < \delta \text{ for all } i = 1, \dots, N\} \quad (41)$$

在模式 A 下：

- $\varepsilon_{\text{avg}} \approx c + O(\delta/\sqrt{N})$ (根据中心极限定理)
- $\text{MAD} \leq 2\delta$ (因为所有值都在彼此 2δ 范围内)

Pattern B (Outlier Pattern): 少数几个大的离群值，其余值接近零。例如：

- k 个值满足 $\varepsilon_i \approx Nc/k$ (大的离群值)
- $N - k$ 个值满足 $\varepsilon_i \approx 0$

在模式 B 下:

- $\varepsilon_{\text{avg}} \approx c$
- $\text{MAD} \approx c(1 - 1/N)$ (由于离群值而很大)

Probability of Pattern A: 对于一个单独的噪声项落入 $(c - \delta, c + \delta)$ 的情况:

$$P(|\varepsilon_i - c| < \delta) = \Phi\left(\frac{c + \delta}{\sigma}\right) - \Phi\left(\frac{c - \delta}{\sigma}\right) \quad (42)$$

对小 δ 使用泰勒展开:

$$P(|\varepsilon_i - c| < \delta) \approx \phi\left(\frac{c}{\sigma}\right) \cdot \frac{2\delta}{\sigma} = \frac{2\delta}{\sigma\sqrt{2\pi}} \exp\left(-\frac{c^2}{2\sigma^2}\right) \quad (43)$$

对于所有 N 个噪声项独立满足此条件:

$$P(\text{Pattern A}) = \left[\frac{2\delta}{\sigma\sqrt{2\pi}} \exp\left(-\frac{c^2}{2\sigma^2}\right) \right]^N \quad (44)$$

Probability of $\varepsilon_{\text{avg}} > c$:

由于 $\varepsilon_{\text{avg}} \sim N(0, \sigma^2/N)$:

$$P(\varepsilon_{\text{avg}} > c) = 1 - \Phi\left(\frac{c\sqrt{N}}{\sigma}\right) \quad (45)$$

对大参数使用米尔比率近似:

$$P(\varepsilon_{\text{avg}} > c) \approx \frac{\sigma}{c\sqrt{2\pi N}} \exp\left(-\frac{c^2 N}{2\sigma^2}\right) \quad (46)$$

关键见解在于, 模式 A 是唯一能使 MAD 保持足够小以满足 $\text{MAD} < (\varepsilon_{\text{avg}} - c)/\alpha$ 的模式。

对于模式 B 和其他由离群值驱动的模式, $\text{MAD} = O(c)$, 而

在给定 $\varepsilon_{\text{avg}} \approx c$ 的条件下 $\varepsilon_{\text{avg}} - c = O(1/\sqrt{N})$ 。因此:

$$\text{MAD} \gg \frac{\varepsilon_{\text{avg}} - c}{\alpha} \quad (47)$$

因此, 改进因子主要由模式 A 主导:

$$\rho \lesssim \frac{P(\text{Pattern A})}{P(\varepsilon_{\text{avg}} > c)} \quad (48)$$

代入表达式:

$$\rho \lesssim \frac{\left[\frac{2\delta}{\sigma\sqrt{2\pi}} \exp\left(-\frac{c^2}{2\sigma^2}\right) \right]^N}{\frac{\sigma}{c\sqrt{2\pi N}} \exp\left(-\frac{c^2 N}{2\sigma^2}\right)} \quad (49)$$

简化:

$$\rho \lesssim \frac{c\sqrt{N}}{\sigma} \left(\frac{2\delta}{\sigma\sqrt{2\pi}} \right)^N \exp\left(-\frac{c^2 N}{2\sigma^2} + \frac{c^2 N}{2\sigma^2}\right) \quad (50)$$

$$\rho \lesssim c\sqrt{N} \left(\frac{2\delta}{\sigma^2\sqrt{2\pi}} \right)^N \quad (51)$$

对于 $\delta = O(\sigma)$, 令 $\delta = k\sigma$, 其中 k 为常数。则:

$$\rho \lesssim c\sqrt{N} \left(\frac{2k}{\sqrt{2\pi}} \right)^N \quad (52)$$

当 $k < \sqrt{\pi/2}$ 时, 项 $(2k/\sqrt{2\pi})^N$ 呈指数衰减。多项式因子 $\sqrt{N}$ 被指数衰减所主导, 从而得到:

$$\rho = O(\sqrt{N} \cdot \exp(-\beta N)) = O(\exp(-\Theta(N))) \quad (53)$$

对于某个正常数 β 。

我们已经建立:

$$P_{\text{FP}}(\text{MAD}) = \rho \cdot P_{\text{FP}}(\text{Avg}) \quad (54)$$

其中 $\rho = O(\exp(-\Theta(N)))$ 随提示数量 N 呈指数衰减。

这证明了 MAD 校准策略相对于简单平均方法提供了指数级的改进。□

结论: MAD 惩罚项有效地 filters out the more probable outlier patterns while only allowing the exponentially rare coherent patterns to trigger false positives, 从而实现了提示诱导噪声的卓越鲁棒性。

A.2 Proof of the Expected Value of MAD

A.2.1 Theorem 1

对于一个随机变量 $X \sim \mathcal{N}(\mu, \sigma^2)$, 其 $\mathcal{C}_{\text{MAD}}$ 的期望值为:

$$\mathbb{E}[\mathcal{C}_{\text{MAD}}] = \sigma \sqrt{\frac{2}{\pi}} \approx 0.8\sigma \quad (55)$$

A.2.2 Proof

Definition. 随机变量 X 的均值绝对偏差, 其均值为 μ , 定义为:

$$\mathcal{C}_{\text{MAD}} = \mathbb{E}[|X - \mu|] \quad (56)$$

令 $X \sim \mathcal{N}(\mu, \sigma^2)$ 。考虑标准化随机变量:

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1) \quad (57)$$

因此:

$$|X - \mu| = \sigma|Z| \quad (58)$$

对两边取期望:

$$\mathbb{E}[|X - \mu|] = \sigma \cdot \mathbb{E}[|Z|] \quad (59)$$

对于具有概率密度函数 $\phi(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2}$ 的 $Z \sim \mathcal{N}(0, 1)$:

$$\mathbb{E}[|Z|] = \int_{-\infty}^{\infty} |z| \cdot \phi(z) dz \quad (60)$$

由于标准正态分布关于零对称以及 $|z|$ 的偶函数性质:

$$\mathbb{E}[|Z|] = 2 \int_0^{\infty} z \cdot \phi(z) dz = \frac{2}{\sqrt{2\pi}} \int_0^{\infty} z \cdot e^{-z^2/2} dz \quad (61)$$

令 $u = z^2/2$, 则 $du = z dz$ 。当 $z = 0$ 时, $u = 0$; 当 $z \rightarrow \infty$ 时, $u \rightarrow \infty$ 。

$$\int_0^\infty z \cdot e^{-z^2/2} dz = \int_0^\infty e^{-u} du = [-e^{-u}]_0^\infty = 1 \quad (62)$$

代回:

$$\mathbb{E}[|Z|] = \frac{2}{\sqrt{2\pi}} \cdot 1 = \sqrt{\frac{2}{\pi}} \quad (63)$$

因此:

$$\mathcal{C}_{\text{MAD}} = \mathbb{E}[|X - \mu|] = \sigma \cdot \mathbb{E}[|Z|] = \sigma \sqrt{\frac{2}{\pi}} \approx 0.8\sigma \quad (64)$$

Algorithm 1 Dynamic Early Exit in Reasoning (DEER)

```

1: Initialization: Large Reasoning Language Model  $\text{LRM}(\cdot)$ , zero-shot-CoT  $zs\_cot$ ,
   question, answer inducer prompt  $\mathbf{I}$ , set of action transition points  $\mathbb{P}$ , end-of-thinking
   delimiter  $\langle /think \rangle$ , maximum length  $max\_len$ , and confidence threshold  $\lambda$ .
2:  $\mathbf{x} \leftarrow zs\_cot + \text{question}$ ,  $\mathbf{r} \leftarrow []$ 
3: while  $len(\mathbf{x}) < max\_len$  do
4:    $y \leftarrow \text{LRM}(\mathbf{x})$ 
5:   if  $y \in \mathbb{P}$  then ▷ Generate thoughts until meets action transition points
6:      $\mathbf{A} \leftarrow \text{LRM}(\mathbf{x} + \mathbf{I})$  ▷ Prompt LRM to generate trial answer tokens
7:     Get  $\mathcal{C}$  according to Equation 5 ▷ Calculate the confidence of the trial answer
8:     if  $\mathcal{C} > \lambda$  then
9:        $\mathbf{x} \leftarrow \mathbf{x} + \langle /think \rangle$ ,  $\mathbf{r} \leftarrow \mathbf{r} + \langle /think \rangle$  ▷ Exit when thinking is sufficient
10:    end if
11:  else
12:     $\mathbf{x} \leftarrow \mathbf{x} + y$ ,  $\mathbf{r} \leftarrow \mathbf{r} + y$ 
13:  end if
14: end while
15: return  $\mathbf{r}$ 
    
```

B More Experiment Setup

Metrics. DEER 的目标是在保持 LRM 正确性性能的同时, 避免因过度思考 (overthinking) 导致的冗余令牌开销。为此, 我们选择 Accuracy (ACC) 和 Generation Length (LEN) 作为评估指标。Accuracy (ACC) 的计算方式如下: $\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}\{\mathcal{M}(\mathcal{LRM}(x_i)) = y_i\}$, 其中 x_i 是问题, y_i 是数据集中的真实答案。 $\mathcal{M}(\cdot)$ 从 LRM 的响应中提取答案。 $\mathbb{I}\{\cdot\}$ 是一个指示函数, 用于判断内部给定条件是否有效。准确性评估基于 Ye et al. (2025) (LIMO) 公开发布的评估框架。直观上, 生成的文本越长, LRM 的推理成本就越高。因此, 我们计算每个样本的平均生成令牌数以评估成本, 如下所示: $\text{Generation Length}(\text{LEN}) = \frac{1}{N} \sum_{i=1}^N |\mathcal{LRM}(x_i)|$, 其中 $|\cdot|$ 测量生成的令牌数量。对于两个编程基准测试, 我们使用 Pass@1 指标来衡量生成代码的正确性。

Implementation details. 所有评估均在零样本思维链 (CoT) 设置下进行, 使用以下提示词: "Please reason step by step, and put your final answer within `\boxed{\}`。" 对于解码策略, 我们在正确性评估中采用贪婪解码和单一样本。我们实验中评估问题的真实答案都是结构良好的数值或选项。因此, 我们直接应用基于规则的评估来验证数学等价性。我们将最大生成长度设置为 16,384,

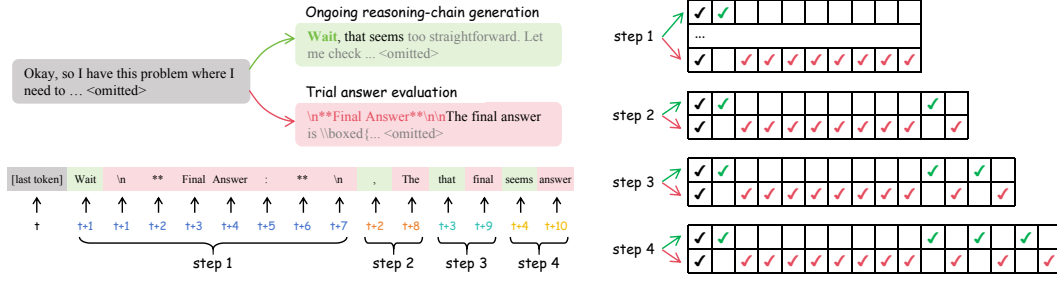


图 9: 分支并行解码与动态 KV 缓存管理。

以确保评估能捕捉到完整的问题解决尝试。对于 DEER，使用的答案诱导提示词是：'\n\n Final Answer\n\n\boxed{' 对于 DEER-Pro，我们额外加入了以下三个提示词：'\n\n Final Answer\n\n\boxed{'、'\n\n Final Answer\n\n\boxed{'、'\n\n Based on the previous thinking, I believe I already know the answer.\n Final Answer\n\n\boxed{'。

C More Method Details

图 9 展示了所提出的分支并行解码加速的工作流程。算法 1 展示了 DEER 的伪代码。

D More Benchmark Descriptions.

Benchmarks. 为了全面评估模型在各种推理能力上的表现，我们选择了 6 个数学推理基准、1 个科学基准和 3 个编码基准，具体如下：

Math Benchmarks:

- **GSM8K** 是一个精心策划的集合，包含 1,319 个初等数学问题。该基准专门用于评估基础数学任务中的多步推理能力。问题通常涉及两到八个连续操作，主要依赖于在多个中间步骤中执行的基本算术。
- **MATH-500** 是一个具有挑战性的基准，包含来自不同高中数学领域（包括预代数、代数和数论）的竞赛级问题。为了与先前研究保持一致，我们采用由 OpenAI 最初策划用于评估的相同 500 个问题子集。
- **AMC 2023** 包含 40 个数学问题，涵盖代数、几何、数论和组合数学。美国数学竞赛（AMC）由美国数学协会（MAA）组织，是一项旨在培养解决问题能力和识别数学人才的著名竞赛。在评估中，我们使用了 LIMO 中来自 AMC 23 的 40 个问题。
- **AIME 2024** 包含从 2024 年美国数学邀请赛（AIME）中选出的 30 个挑战性问题。这项著名的竞赛评估参与者在多个领域的数学推理能力，包括算术、代数、计数、几何、数论、概率以及其他中学数学主题。AIME 的一个显著特点是其答案格式：所有解必须是 000 到 999（含）之间的整数。每个问题根据 Art of Problem Solving（AoPS）的难度等级（1-5）进行分类。除了这三个数学问题，我们还对科学问题进行了评估。

- AIME 2025 包含从 2025 年美国数学邀请赛（AIME）中选出的 30 个挑战性问题。
- OlympiadBench OlympiadBench 是一个奥林匹克级别的双语多模态科学基准数据集，旨在挑战和评估大型语言模型和大型多模态模型的高级能力。它包含 8,476 个源自奥林匹克级别数学和物理竞赛的问题，包括中国高考中的问题。我们的实验评估选择了与 LIMO 中使用的相同的 675 个样本子集，允许对生成的答案进行直接的基于规则的评估。

Science Benchmarks:

- GPQA 是一个博士级别的基准，包含跨越物理、化学和生物学子领域的高质量问題。值得注意的是，在这些领域拥有博士学位的领域专家在该数据集上仅达到 69.7% 的准确率。在我们的实验中，我们特别选择了最高质量的子集，即 GPQA Diamond（由 198 个问题组成）。

Programming Benchmarks:

- HumanEval 由 OpenAI 提出，包含 164 个手工制作的（为避免数据泄露）专注于基础算法的 Python 编程任务，每个任务都包含函数签名、文档字符串、规范解决方案和单元测试。
- BigCodeBench 被设计为一个面向现实世界的基准测试，包含 1,140 个任务，需要与 139 个库和多样化的函数调用进行交互。
- LiveCodeBench 是一个新提出的基准数据集，旨在评估大语言模型在代码生成及相关任务上的能力。它旨在缓解现有基准中存在的测试集污染等问题，强调代码生成之外的场景，确保高质量的问题来源、充足的测试用例以及平衡的难度级别。该数据集包含从 AtCoder、LeetCode 和 CodeForces 等知名竞赛编程平台收集的问题，这些问题的收集时间窗口是特定的。我们的评估基于 LiveCodeBench-v5，它包含从 2023 年 5 月到 2025 年 1 月收集的 880 个编程问题。

在我们的实验中， $8 \times 80\text{g}$ 内存的 H100 被用于执行评估。

E More LRM Descriptions.

在这项工作中，我们验证了 DEER 在 12 个推理模型上的有效性。评估的模型包括：Qwen3-1.7B、Qwen3-4B、Qwen3-8B、Qwen3-14B、Qwen3-32B、DeepSeek-R1-Distill-Qwen-1.5B、DeepSeek-R1-Distill-Qwen-7B、DeepSeek-R1-Distill-Qwen-14B、DeepSeek-R1-Distill-Qwen-32B、QwQ-32B、DeepSeek-R1-671B 以及 Llama-3.1-Nemotron-Nano-8B-v1。DeepSeek-R1-Distill 系列中的所有模型均使用 DeepSeek-R1 模型生成的推理数据进行监督微调。Qwen3-1.7B、Qwen3-4B、Qwen3-8B 和 Qwen3-14B 模型采用一种称为“强到弱蒸馏”的方法进行训练。通过强化学习训练的非蒸馏模型 QwQ-32B 和 Qwen3-32B 在推理基准测试中表现出具有竞争力的性能，与 DeepSeek-R1-671B 相当。由于计算资源限制，我们基于 KTransformers (kvcache ai, 2025) 实现了 Deepseek-R1 的量化版本。

F Computational Cost Analysis

在本节中，我们提供一个理论分析来证明 DEER 能有效降低计算成本。令 L 表示原始 CoT 方法生成的总长度， α 代表 DEER 相对于 L 的压缩比，使得 DEER 生成长度为 αL 的序列。然后，我们定义 k 为这 αL 个推理标记内的答案诱导触发次数， m 为每次答案诱导的平均生成长度，这通常是一个较小的常数。在 Transformer 推理过程中，主要的计算开销源于注意力计算，这是我们的主要关注点。假设生成过程采用了键值缓存技术，每个新标记只需与缓存的键值对计算注意力。

F.1 Computational Cost Analysis on Time

对于原始的 CoT 方法，其计算成本为：

$$T = O(1) + O(2) + \dots + O(L) = O(L^2) \quad (65)$$

对于我们的 DEER 方法，计算成本包含两个部分：主推理链中的 αL 次前向传递以及答案诱导所需的 km 次前向传递。

首先，我们计算主推理链的成本：

$$T_{\text{main}} = \sum_{t=1}^{\alpha L} t = \frac{\alpha L(\alpha L + 1)}{2} = O(\alpha^2 L^2) \quad (66)$$

接下来，对于答案诱导期间的计算开销，我们首先计算单次诱导的时间成本。假设第 j 次答案诱导在位置 p_j 处被触发，其成本为：

$$C_{\text{single}} = \sum_{i=1}^m (p_j + i - 1) = m \cdot p_j + \sum_{i=1}^m (i - 1) = m \cdot p_j + \frac{m(m-1)}{2} \quad (67)$$

假设诱导位置 p_j 在区间 $[0, \alpha L]$ 上均匀分布，则平均诱导位置为 $E[p_j] \approx \frac{\alpha L}{2}$ 。因此，答案诱导的总成本为：

$$C_{\text{induce}} \approx k \cdot m \cdot \frac{\alpha L}{2} + k \cdot \frac{m(m-1)}{2} = O(k \cdot m \cdot \alpha L) + O(k \cdot m^2) \quad (68)$$

即使在最坏情况下，即大多数触发点 p_j 聚集在推理过程末尾附近，平均诱导位置为 $E[p_j] \approx \alpha L$ 。此时答案诱导的总成本为：

$$C_{\text{induce}} \approx k \cdot m \cdot \alpha L + k \cdot \frac{m(m-1)}{2} = O(k \cdot m \cdot \alpha L) + O(k \cdot m^2) \quad (69)$$

由于每次答案诱导的长度 m 与推理长度 L 相比可以忽略不计，我们有：

$$C_{\text{induce}} \approx O(k \cdot m \cdot \alpha L) \quad (70)$$

最后，DEER 的总成本为：

$$C_{\text{DEER}} = C_{\text{main}} + C_{\text{induce}} = O(\alpha^2 L^2) + O(k \cdot m \cdot \alpha L) \quad (71)$$

DEER 将二次项从 $O(L^2)$ 减少到 $O(\alpha^2 L^2)$ ，同时仅引入了一个线性项 $O(k \cdot m \cdot \alpha L)$ 。由于在长链式推理中 $k, m \ll L$ ，二次项减少所带来的节省远超过额外线性项的开销。这一分析有效地解释了第 4.4 节中观察到的超线性加速现象。

F.2 Computational Cost Analysis on Memory

内存开销分析可以分解为两个组成部分：来自 KV 缓存的主要内存消耗，以及来自并行解码操作的额外开销。

Peak Memory Reduction. 现代 LLM 推理中的主要内存开销源于注意力键和值在 KV 缓存中的存储，其大小与处理的序列长度成线性比例。标准的思维链方法需要为所有 L 个令牌维护 KV 缓存，导致内存复杂度为 $O(L)$ 。通过在位置 αL 处提前终止（其中 $\alpha < 1$ ），DEER 有效地将推理期间的峰值序列长度从 L 减少到 αL 。因此，峰值 KV 缓存内存消耗从 $O(L)$ 减少到 $O(\alpha L)$ 。

这种内存减少在处理长上下文推理任务时尤其有价值。除了降低单个请求的峰值内存需求外，这种方法使系统能够在批处理期间，在相同的内存约束下容纳更多的并发请求，从而提高整体吞吐量。

Additional Overhead for Parallel Decoding. 所提出的并行解码变体，在执行答案归纳前向传播的同时进行主要推理，引入了极少的额外内存开销。这种效率是通过在现代推理框架（如 vLLM）中实现的前缀缓存和共享机制实现的。当多个推理分支共享一个共同的前缀序列时，它们 KV 缓存的相应部分通过诸如 vLLM 的 PagedAttention 等技术，在物理内存中仅需单次存储。

在并行解码期间，答案归纳分支几乎不会产生额外的 KV 缓存开销，因为它完全利用了主推理分支已经计算好的 KV 缓存。唯一的边际额外内存需求来自于存储代表答案归纳分支解码状态的有限数量的令牌。

对于代码生成任务，我们的实现包含了特定的优化，即仅生成前 50 个令牌用于置信度估计。这一设计选择是一个实现细节，而非核心方法学贡献。实验验证证实，利用部分答案令牌进行提前退出的置信度计算对于编码任务仍然有效。

G Investigation of Reasoning Transition Monitors

在正文的 ?? 节中，我们的实验表明，推理转换监控器的选择对 DEER 产生微妙的影响，主要体现在潜在提前退出机会的数量如何影响最终生成长度。在本节中，我们研究基于语言标记和基于熵的监控方法之间的潜在联系。

表 10 展示了在多个数据集和模型上，语言标记与其他标记之间平均令牌熵的对比分析。我们的发现表明，语言标记的熵显著高于其他标记，这表明基于语言标记的方法本质上针对的是存在多个候选动作的高熵位置。

此外，我们计算了最终层隐藏状态表示中连续令牌之间的余弦相似度得分，将语言标记与其相邻令牌的相似度与常规令牌对进行比较。该相似度指标作为模型推理连贯性的指示器：高相似度反映连续、连贯的推理过程，而低相似度则标志着推理转换的发生。表 11 中呈现的结果表明，语言标记的相似度得分显著更低，这表明表征连续性出现了中断。

总的来说，这些实验提供了有力的证据，表明大型推理语言模型并非在静默中经历不确定状态；相反，它们通过语言明确地表达不确定性。DEER 所利用的外部语言标记构成了内部状态转换的直接表现，从而为我们的方法理论基础提供了强有力的实证支持。

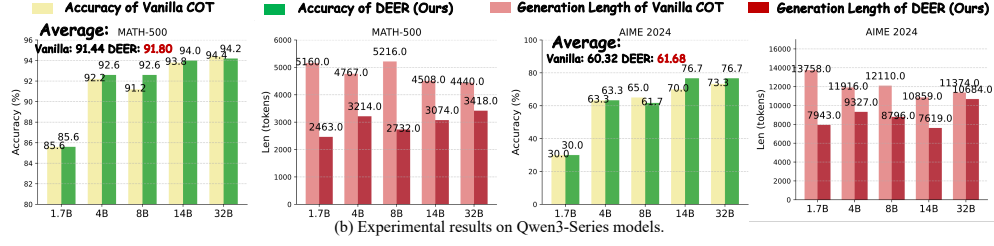


图 10: DEER 与 Vanilla CoT 在不同规模的 Qwen3 系列模型上, 在 MATH-500 和 AIME 2024 数据集上的实验结果对比。

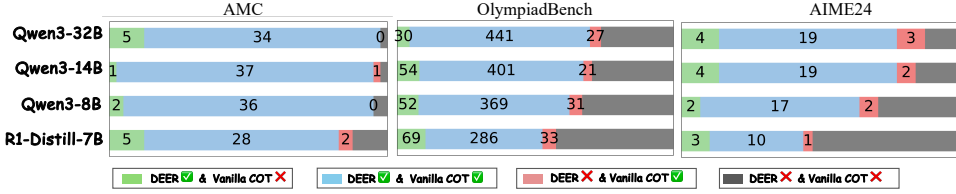


图 11: DEER 与 Vanilla CoT 更详细的实验结果对比。✓ 表示正确答案, ✗ 表示错误答案。

H Investigation into the Reasons Behind DEER's Threshold Robustness

在正文第??节中, 我们通过在模型和数据集上的实验, 证明了 DEER 对阈值 λ 的鲁棒性。在本节中, 我们探究这种鲁棒性的根本来源。我们分析了三个模型在三个数学推理数据集上所有潜在退出位置处诱导答案的置信度分数, 计算了分数落在三个不同区间的比例。具体来说, 0-0.9 代表低置信度区间, 0.97-1.0 代表高置信度区间, 而 0.9-0.97 则构成了易出错的灰色地带。

表12中呈现的结果表明, 模型的置信度分布呈现出明显的极化现象。绝大多数情况集中在高度自信或信心不足的两个极端, 中间范围 (易出错的灰色地带) 的存在极少。当我们的方法诱导模型生成最终答案时, 置信度分数遵循独特的 U 形分布, 在 0.9 到 0.97 之间的概率质量极低。这一现象表明, 当模型基于其先前的推理链对某个答案拥有足够把握时, 它会以极高的概率 (通常超过 0.99) 生成该答案。反之, 当存在不确定性时, 分配的概率则会大幅下降。

此外, 我们观察到, 在较简单的问题 (GSM8K) 上, 所有三个模型的高置信度分数比例都高于低置信度分数比例; 而在更具挑战性的问题 (AIME24) 上, 低置信度分数的比例则超过了高置信度分数。这一观察进一步验证了 DEER 方法的合理性: 分配给试验答案的置信度准确地反映了现有推理是否足以解决问题。因此, 在困难问题上置信度通常较低, 导致在初始阶段出现许多失败的早期退出尝试。这种模式也解释了 DEER 在不同问题难度上表现各异的原因。在较简单的问题上, 模型表现出足够的信心, 从而获得更好的压缩效果。在具有挑战性的问题上, 模型变得更加谨慎, 导致压缩效果较弱, 但仍能保持令人满意的准确性。这种自适应行为表明, DEER 能够根据问题复杂度, 自然地平衡计算效率和求解质量。

More experiments across Model Sizes on Qwen3. 图 10 中 Qwen3 系列模型在不同模型规模和推理难度上的表现, 与第 4.2 节中提出的发现是一致的。

表 2: 比较多种模型和数据集上的 Vanilla、DEER 和 DEER-PRo。Acc = 准确率 (%), Len = 平均词元数, CR = 压缩比。

Method	GSM8K			MATH			AMC			AIME			GPQA			Overall	
	Acc	Len	CR	Acc	Len	CR	Acc	Len	CR	Acc	Len	CR	Acc	Len	CR	Acc	CR
DeepSeek-R1-Distill-Qwen-1.5B																	
Vanilla	76.1	1,617	100%	69.0	6,018	100%	52.5	8,819	100%	23.3	13,702	100%	7.1	13,029	100%	45.6	100%
DEER	74.7	984	60.9%	67.8	2,497	41.5%	60.0	5,496	62.3%	23.3	9,557	69.7%	12.1	5,762	44.2%	47.6	55.7%
DEER-PRo	77.3	1,062	65.7%	70.0	2,891	48.0%	62.5	5,701	64.6%	26.7	10,390	75.8%	14.5	6,820	52.3%	50.2	61.3%
Qwen3-4B																	
Vanilla	94.1	2,175	100%	92.2	4,767	100%	87.5	7,443	100%	63.3	11,916	100%	46.5	9,294	100%	76.7	100%
DEER	94.5	1,250	57.5%	92.6	3,214	67.4%	87.5	4,906	65.9%	63.3	9,327	78.3%	47.5	3,275	35.2%	77.1	60.9%
DEER-PRo	94.5	1,301	59.8%	93.0	3,517	73.8%	92.5	5,153	69.2%	65.0	9,651	81.0%	49.2	3,750	40.3%	78.8	64.8%
Qwen3-1.7B																	
Vanilla	90.1	2,045	100%	85.6	5,160	100%	70.0	8,637	100%	30.0	13,758	100%	35.9	9,271	100%	62.3	100%
DEER	90.3	1,066	52.1%	85.6	2,463	47.7%	70.0	4,673	54.1%	30.0	7,943	57.7%	43.4	3,549	38.3%	63.9	50.0%
DEER-PRo	90.7	1,261	61.7%	87.2	2,702	52.4%	75.0	5,143	59.5%	35.0	8,644	62.8%	44.5	3,960	42.7%	66.5	55.8%

表 3: 编程任务上的实验结果。Acc = 准确率 (%), Tok. = 平均令牌数, CR = 压缩率。

Model	Method	HumanEval			BigCodeBench			LiveCodeBench			Overall	
		Acc	Tok.	CR	Acc	Tok.	CR	Acc	Tok.	CR	Acc	CR
R1-Distill-Qwen Series												
32B	Vanilla	91.5	3,861	100%	44.5	5,459	100%	56.0	9,109	100%	64.0	100%
	DEER	93.9	1,254	32.5%	46.1	1,929	35.3%	56.6	3,677	40.4%	65.5	36.1%
14B	Vanilla	89.0	4,039	100%	40.9	4,806	100%	52.7	9,259	100%	60.9	100%
	DEER	90.9	1,000	24.8%	40.7	1,583	32.9%	52.1	4,000	43.2%	61.2	33.6%
7B	Vanilla	78.6	5,666	100%	26.1	8,516	100%	38.4	10,482	100%	47.7	100%
	DEER	78.6	913	16.1%	25.2	1,605	18.8%	40.3	2,582	24.6%	48.0	19.9%
Qwen3 Series												
14B	Vanilla	93.3	3,277	100%	44.6	5,072	100%	73.4	8,203	100%	70.4	100%
	DEER	93.9	1,118	34.1%	44.3	792	15.6%	74.1	5,437	66.3%	70.8	38.7%
8B	Vanilla	85.4	3,904	100%	37.9	6,994	100%	64.7	8,871	100%	62.7	100%
	DEER	87.8	793	20.3%	41.1	608	8.7%	65.1	4,257	48.0%	64.7	25.7%
4B	Vanilla	91.5	3,768	100%	36.1	7,804	100%	64.7	8,789	100%	64.1	100%
	DEER	92.7	1,050	27.9%	37.1	826	10.6%	63.5	5,626	64.0%	64.4	34.2%
1.7B	Vanilla	83.5	3,580	100%	25.7	6,151	100%	51.7	9,447	100%	53.6	100%
	DEER	84.1	2,236	62.5%	28.3	2,104	34.2%	51.4	8,425	89.2%	54.6	62.0%

Performance on SoTA Reasoning Models. 我们在两个最先进的推理模型上评估了 DEER 的有效性: Qwen3-32B (代表稠密模型) 和 Deepseek-R1 671B (代表 MoE 模型)。为了充分利用它们的推理能力,我们将其最大序列长度分别设置为官方推荐的 32k 和 16k。最大长度的影响将在下一节进一步讨论。由于计算资源限制,我们基于 KTransformers (kvcache

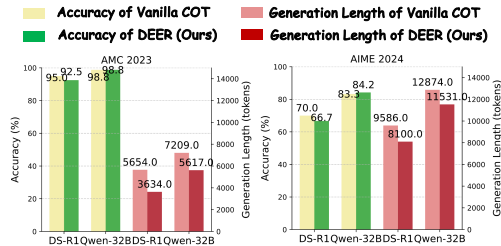


图 12: 在先进模型上的性能表现。

表 4: 更多模型和任务上的额外阈值敏感性实验。

Qwen3-14B	GSM8K	MATH	AMC
Vanilla	95.1	93.8	95.0
0.80	96.0	93.8	93.8
0.85	95.7	94.4	93.8
0.90	96.1	93.8	95.0
0.95	96.0	94.0	95.6
0.97	95.7	93.8	94.4
Qwen3-8B	GSM8K	MATH	AMC
Vanilla	94.9	91.2	87.5
0.80	95.2	91.4	88.8
0.85	94.9	92.0	90.0
0.90	95.5	92.8	91.3
0.95	95.3	93.2	92.5
0.97	95.3	93.0	92.5

ai, 2025) 实现了 Deepseek-R1 的量化版本。图

12 详细展示了 DEER 在两个具有挑战性的数据集 AIME 和 MATH 上的表现。结果表明, DEER 在保持有竞争力的准确率 (R1 在每个数据集上仅多犯一个错误) 的同时, 显著地将序列长度减少了 10.4% - 35.7%。

Performance Trends across More Model Sizes and Benchmarks. 为了更全面地展示 DEER 的有效性, 并方便研究人员进行比较, 我们在七个推理基准测试和十一个大型推理语言模型上呈现了实验结果。图 K 比较了 DEER 与原始思维链的实验结果, 表明论文正文中得出的结论在更多基准测试和额外模型上保持一致。

除了流行的推理模型, 我们也评估了 DEER 在较少使用的模型上的表现。如局限性部分所述, Llama-3.1-Nemotron-Nano-8B-v1 在生成中间答案时始终表现出较低的信心度, 导致其在评估期间的提前停止率显著低于主流模型 (Qwen3-8B: 80%, R1-Distill-Qwen-7B: 85%, Llama-3.1-Nemotron-Nano-8B-v1: 55%)。因此, 如表 5 所示, Llama-3.1-Nemotron-Nano-8B-v1 的推理效率提升有限。尽管如此, DEER 仍然有效地缓解了该模型中的过度思考问题, 这体现在它能够通过提前停止来防止后续推理步骤将正确答案改为错误答案。

Performance with Different Decoding Configurations. 由于 DeepSeek-R1-Distill-Series 模型的配置推荐最大长度为 16k (16,384), 为了保持设置的一致性, 我们在主要实验中也以相同设置评估了 Qwen3-14B。实际上, 这个长度对于大多数现实世界的应用来说已经足够。此外, 为了确保实验的稳定性和可复现性, 我们在主要实验中采用了贪心解码。尽管如此, 为了更全面地评估 DEER 的性能, 我们进一步在 Qwen3 系列模型的更大变体 (8B、14B、32B) 上使用官方推荐的解码策略 ($\max_len = 32768$, $\text{top_p} = 0.95$, $\text{temperature} = 0.6$) 进行了实验。图 8 显示, 在这些配置下, DEER 仍然显著有效, 证明了我们方法的鲁棒性。

Performance with Different Multi-Token Confidence Averaging Methods. 在正文中, 我们提到了采用几何平均策略来计算多标记答案的置信度分数, 因为它更符合语言模型中联合概率计算的乘法性质, 并且对低概率值表现出更高的敏感性。在本节中, 我们补充了使用算术

表 5: 在 MATH-500 (DeepSeek-R1-Distill-Qwen-14B) 上使用不同推理转换监控器的结果。DEER(W) 表示通过 Wait 进行转换, DEER(A) 表示通过 Alternatively 进行转换, DEER(Ent) 表示通过熵阈值进行转换。块大小表示一个推理块 (思考) 的长度 (标记数量), 块数量表示推理块的数量。

Method	Accuracy	Tokens	Chunk Size	Chunk Num	Exit Ratio	Exit Acc.
Vanilla	88.6	3815	—	—	—	—
DEER(W)	89.6 _{+1.0}	2572 _{-32.6%}	259.5	14.7	87.6%	93.4%
DEER(A)	90.8 _{+2.2}	2775 _{-27.3%}	719.8	5.3	54.8%	91.2%
DEER(Ent)	90.2 _{+1.6}	2339 _{-38.7%}	183.7	20.8	90.2%	93.0%

平均计算 (DEER-AM) 的对比实验分析, 采用相同的 0.95 提前退出阈值。表7中的结果表明, 与 DEER-GM 相比, DEER-AM 的准确率显著下降, 同时在压缩比方面仅实现了边际改进。这表明算术平均稀释了低值标记的贡献, 导致整体置信度分数虚高, 进而引发过早的错误退出。因此, 我们建议使用几何平均来估计真实的置信度分数。

Error Bars with 95% Confidence Intervals. 为了证明 DEER 准确率提升的统计显著性, 我们在两个模型上进行了多次实验运行, 并计算了 95% 置信区间的误差条。具体来说, 我们在 GSM8K、MATH 和 GPQA 基准上进行了四次独立运行。鉴于 AMC23 和 AIME24 的样本量有限, 我们将这些数据集的实验重复次数增加到八次。表9中呈现的结果证实, 我们的方法所实现的准确率提升具有统计显著性。

I Case Study Details

在图 13 中, 我们提供了在 MATH-500 数据集上的结果示例, 以直观展示 DEER 的有效性。DEER 的设计确保其在提前退出之前遵循与原始思维链方法相同的推理过程。如绿色框所示, 两种方法都在第一个推理步骤中得出了正确答案。区别在于, 我们的方法在评估试验答案的置信度足够高后提前退出, 从而产生正确结果。相比之下, 原始思维链方法继续进行下一个推理动作。经过双重检查并切换推理方法后, 由于两种方法得出的答案不一致, 模型陷入了无尽的验证循环, 最终未能提供最终答案。此外, 图 14、15、16 提供了额外的生成示例, 以更全面地展示 DEER 提前退出机制的有效性, 并说明该方法的底层机制。

图 17 展示了 DEER 应用于一个数学示例的详细过程。可以观察到, 在每个推理切换点 (“Wait” 标记处), DEER 会生成一个试验答案并评估其置信度。置信度的变化与当前推理块和试验答案的可靠性是一致的。这表明大型推理模型隐式地知道何时提前退出, 而我们的方法简单地实现了模型自身的这种潜力。

J Related Work Details

Open-AI o1 (OpenAI, 2025) 的出现, 确立了测试时缩放 (Snell et al., 2024) 作为 LLM 社区的一个关键研究方向。这种方法增强了 LLM 的慢思考能力, 使其在解决复杂问题方面取得突破。近期 DeepSeek-R1 (DeepSeek-AI et al., 2025) 的开源进一步激发了人们对本地部署推理模型的兴趣。然而, 两个关键挑战已经浮现: 1) 生成过长的思维链会显著降低推理效率, 2) 越来越多的实证证据 (Chen et al., 2025b; Team et al., 2025a) 揭示了它们容易陷入

表 6: Vanilla、DEER-W 和 DEER-Ent 在多个模型和数据集上的比较。Acc = 准确率 (%), Len = 平均标记数, CR = 压缩比。

Method	GSM8K			MATH			AMC			AIME			GPQA			Overall	
	Acc	Len	CR	Acc	Len	CR	Acc	Len	CR	Acc	Len	CR	Acc	Len	CR	Acc	CR
DeepSeek-R1-Distill-Qwen-7B																	
Vanilla	89.6	1,484	100%	87.4	3,858	100%	78.8	6,792	100%	41.7	13,765	100%	23.7	10,247	100%	64.2	100%
DEER-W	90.6	917	61.8%	89.8	2,143	55.5%	85.0	4,451	65.5%	49.2	9,839	71.5%	31.3	5,469	53.4%	69.2	61.5%
DEER-Ent	90.8	876	59.0%	89.2	2,261	58.6%	85.0	4,072	60.0%	48.4	8,961	65.1%	29.6	5,037	49.2%	68.6	58.4%
Qwen3-14B																	
Vanilla	95.1	2,047	100%	93.8	4,508	100%	95.0	7,139	100%	70.0	10,859	100%	60.1	7,339	100%	82.8	100%
DEER-W	95.3	840	41.0%	94.0	3,074	68.2%	95.0	4,763	66.7%	76.7	7,619	70.2%	57.6	2,898	39.5%	83.7	57.1%
DEER-Ent	96.1	803	39.2%	93.8	2,979	66.1%	93.3	4,903	68.7%	73.3	7,128	65.6%	58.1	2,818	38.4%	82.9	55.6%
Qwen3-8B																	
Vanilla	94.9	2,245	100%	91.2	5,216	100%	87.5	7,986	100%	65.0	12,110	100%	51.5	9,145	100%	78.0	100%
DEER-W	95.2	1,071	47.7%	92.6	2,732	52.4%	92.5	4,392	55.0%	61.7	8,796	72.6%	52.5	3,111	34.0%	78.9	52.3%
DEER-Ent	95.8	1,037	46.2%	93.6	2,789	53.5%	91.3	4,003	50.1%	63.3	8,328	68.8%	51.5	3,248	35.5%	79.1	50.8%

表 7: 比较多个模型和数据集上的 Vanilla、DEER-GM（几何平均）和 DEER-AM（算术平均）。Acc = 准确率 (%), Len = 平均词元数, CR = 压缩比。

Method	GSM8K			MATH			AMC			AIME			GPQA			Overall	
	Acc	Len	CR	Acc	Len	CR	Acc	Len	CR	Acc	Len	CR	Acc	Len	CR	Acc	CR
DeepSeek-R1-Distill-Qwen-7B																	
Vanilla	89.6	1,484	100%	87.4	3,858	100%	78.8	6,792	100%	41.7	13,765	100%	23.7	10,247	100%	64.2	100%
DEER-GM	90.6	917	61.8%	89.8	2,143	55.5%	85.0	4,451	65.5%	49.2	9,839	71.5%	31.3	5,469	53.4%	69.2	61.5%
DEER-AM	90.2	832	56.1%	88.2	1,879	48.7%	80.0	3,872	57.0%	43.3	8,095	58.8%	22.6	4,116	40.2%	64.9	52.2%
Qwen3-14B																	
Vanilla	95.1	2,047	100%	93.8	4,508	100%	95.0	7,139	100%	70.0	10,859	100%	60.1	7,339	100%	82.8	100%
DEER-GM	95.3	840	41.0%	94.0	3,074	68.2%	95.0	4,763	66.7%	76.7	7,619	70.2%	57.6	2,898	39.5%	83.7	57.1%
DEER-AM	95.3	811	39.6%	92.4	2,620	58.1%	90.0	4,513	63.2%	63.3	6,933	63.8%	53.6	2,508	34.2%	78.9	51.8%
Qwen3-8B																	
Vanilla	94.9	2,245	100%	91.2	5,216	100%	87.5	7,986	100%	65.0	12,110	100%	51.5	9,145	100%	78.0	100%
DEER-GM	95.2	1,071	47.7%	92.6	2,732	52.4%	92.5	4,392	55.0%	61.7	8,796	72.6%	52.5	3,111	34.0%	78.9	52.3%
DEER-AM	94.9	972	43.3%	92.0	2,522	48.4%	87.5	3,899	48.8%	56.7	7,697	63.6%	49.7	2,950	32.3%	76.2	47.3%

过度思考——即模型在达到最优输出点后仍继续推理的现象。Zhang et al. (2025c) 引入了一个名为 S1-Bench 的新基准，用于测试 LRM 在简单任务上的性能，评估这些 LRM 的过度思考问题。遵循 (Sui et al., 2025; Wang et al., 2025a) 中建立的高效推理分类法，我们将相关工作分为三类：基于后训练、基于提示和基于输出的高效推理方法。Post-training based efficient reasoning 方法使用监督微调 (Yu et al., 2024; Kang et al., 2025; Xia et al., 2025; Ma et al., 2025b; Munkhbat et al., 2025; Zhu et al., 2025; Liu et al., 2024; Han et al., 2024; Qiao et al., 2025; Yu et al., 2025) 配合可变长度的思维链数据，或在强化学习中引入长度奖励 (Team et al., 2025b; Luo et al., 2025a; Aggarwal & Welleck, 2025; Arora & Zanette, 2025; Yeo et al., 2025; Shen et al., 2025b; Qu et al., 2025; Cui et al., 2025; Dai et al., 2025; Liu et al., 2025a;b; Tu et al., 2025; Wang et al., 2025c; Dumitru et al., 2025; Li et al., 2025a; Jiang et al., 2025a; Zhang et al., 2025b)，以使模型能够自适应地生成不同长度的思维链。然而，这些方法通常需要大量的计算资源，并且在数据集构建方面面临挑战。最近，一些工作 (Hao et al., 2024; Shen et al., 2025c; Cheng & Van Durme, 2024; Dang et al., 2025; Shen et al., 2025a; Su et al., 2025; Tan et al., 2025; Saunshi et al., 2025; Zhang et al., 2025d) 表

表 8: 在官方推荐设置（最大 _ 长度 = 32768, top_p = 0.95, 温度 = 0.6）下, Qwen3 系列模型的实验结果。

Budget	Method	GSM8K		MATH-500		AMC23		AIME24		AIME25		OlympiadB		Overall	
		Acc	Tok.	Acc	Tok.	Acc	Tok.	Acc	Tok.	Acc	Tok.	Acc	Tok.	Acc	CR
Qwen3-8B															
32k	Vanilla	95.7	2246	93.8	5368	93.8	9424	70.0	16717	65.0	17880	67.9	11025	81.0	100%
	DEER	95.5	981	94.0	3227	95.0	5898	75.8	12465	63.3	15135	67.0	9075	81.8	68.0%
Qwen3-14B															
32k	Vanilla	95.7	1699	94.8	4800	95.0	6837	75.0	14347	76.7	16437	68.7	9992	84.3	100%
	DEER	95.8	933	95.0	3301	96.3	6299	74.2	10896	76.7	15014	68.9	8263	84.5	77.6%
Qwen3-32B															
32k	Vanilla	96.0	1714	95.8	4609	98.8	7209	83.3	12874	78.3	15292	69.3	9775	86.9	100%
	DEER	95.8	992	95.4	3325	98.8	5617	84.2	11531	78.3	13981	69.8	8671	87.1	79.6%

表 9: 在推理基准测试上的准确率表现, 包含 95% 置信区间。

Model	GSM8K	MATH	AMC23	AIME24	GPQA
Vanilla (ds-7B)	0.897 [0.891, 0.902]	0.877 [0.869, 0.884]	0.794 [0.767, 0.821]	0.425 [0.400, 0.449]	0.247 [0.203, 0.291]
DEER (ds-7B)	0.904 [0.896, 0.912]	0.897 [0.883, 0.911]	0.856 [0.835, 0.878]	0.492 [0.463, 0.520]	0.299 [0.257, 0.341]
Vanilla (Qwen3-14B)	0.948 [0.942, 0.955]	0.938 [0.932, 0.943]	0.938 [0.918, 0.957]	0.708 [0.660, 0.757]	0.596 [0.571, 0.621]
DEER (Qwen3-14B)	0.955 [0.949, 0.962]	0.942 [0.936, 0.948]	0.953 [0.940, 0.967]	0.754 [0.718, 0.791]	0.587 [0.566, 0.608]

明, 使用潜在表征来替代显式的文本推理步骤, 可以使推理模型更加高效。然而, 此类方法通常需要在精心策划的数据集上进行多轮次的监督微调 (Hao et al., 2024; Xu et al., 2025d), 这会导致模型对输出格式的过拟合, 从而损害其固有的表达能力和泛化能力。

Prompt-based efficient reasoning 方法 (Han et al., 2024; Xu et al., 2025b; Lee et al., 2025; Renze & Guven, 2024; Chen et al., 2024) 使用不同的提示词来强制推理模型生成更简洁的思维链, 减少不必要的推理步骤。特别是, (Aytes et al., 2025; Chuang et al., 2024; 2025; Ong et al.) 根据查询的难度为其分配不同的提示词, 从而调整推理模型生成的思维链长度。我们也在表 1 中探索了我们的方法与提示设计相结合的性能, 证明了在保持相当准确性的同时, 可以进一步减少推理链的长度。大多数 Output-based efficient reasoning 方法侧重于优化大型语言模型的最佳 N 采样, 例如剪枝低质量样本 (Xie et al., 2023; Liao et al., 2025) 以及在多个样本达到自治时实施提前停止 (Li et al., 2024; Manvi et al., 2024; Aggarwal et al., 2023)。然而, 随着如 R1 等先进推理模型的引入, 对最佳 N 采样方法的依赖有所减少, 因为这些模型本身就展现出强大的独立推理能力。最近, 两项同时进行的工作与我们的动机相似。Zhang et al. (2025a) 也提出基于试验答案提前终止, 但需要一个额外的探针模型来判断正确性。他们侧重于增强探针模型的验证能力, 而我们的方法探索如何使模型能够自我决定何时提前退出并无缝集成到现有的推理逻辑中。Ma et al. (2025a) 提示推理模型在解码过程中直接输出最终答案, 但仅在与基线相比的低预算场景下或适配到最佳 N 方法时取得更好性能, 这限制了其适用性和泛化能力。Song et al. (2025) 通过保留获得高重要性分数的 KV 缓存来周期性压缩 KV 缓存, 利用推理路径的语义稀疏性来加速推理。Jiang et al. (2025b) 使用教师模型执行技能感知的步骤分解和内容剪枝, 然后将剪枝后的推理路径蒸馏到学生模型中。Huang et al. (2025) 将引导方向投影到低维激活流形上, 并通过干预激活来减少思考令牌。

表 10: Linguistic Markers 与 Other Tokens 的标记熵。

Qwen3-8B	Linguistic Markers	Other Tokens
gsm8k	0.901	0.438
math	1.058	0.385
gpqa	1.269	0.500
DS-7B	Linguistic Markers	Other Tokens
gsm8k	1.550	0.658
math	1.753	0.565
gpqa	1.241	0.510

表 11: Linguistic Markers 与 Other Tokens 之间的隐藏状态余弦相似度。

Qwen3-8B	Linguistic Markers	Other Tokens
gsm8k	0.262	0.543
math	0.237	0.493
gpqa	0.240	0.509
DS-7B	Linguistic Markers	Other Tokens
gsm8k	0.306	0.608
math	0.247	0.530
gpqa	0.231	0.505

K Use of LLMs

在准备本稿件的过程中，大型语言模型（LLMs）被用作语言润色的辅助工具。在稿件准备的最后阶段，我们利用 LLMs 对选定的段落进行了语言上的精炼，包括语法检查、句子结构优化和表达标准化。此过程仅限于语言层面的改进，不涉及任何实质性学术内容的生成或修改，包括研究见解、数据分析或结论推导。需要强调的是，本文中提出的所有核心论点、研究方法、实验设计、数据分析和结论均由作者独立完成。LLMs 仅作为语言处理辅助工具，作者对所有内容承担完全的学术责任。

表 12: 不同模型在各任务上的置信区间分布 (%)。

Qwen3-14B	0–0.9	0.9–0.97	0.97–1.0
gsm8k	38.92	5.95	55.06
math	49.53	4.83	45.46
aime	77.20	2.45	20.35
Qwen3-8B	0–0.9	0.9–0.97	0.97–1.0
gsm8k	35.31	5.34	59.12
math	45.27	4.58	49.98
aime	78.61	2.39	19.00
DS-7B	0–0.9	0.9–0.97	0.97–1.0
gsm8k	26.54	5.29	68.17
math	34.09	5.46	60.45
aime	80.90	1.41	17.69

Question:		A Senate committee has 5 Democrats, 5 Republicans, and 1 Independent. In how many ways can they sit around a circular table if all the members of each party all sit next to each other? (Two seatings are considered equivalent if one is a rotation of the other.)	
Answer:		DEER	Vanilla CoT
[Slow thinking] Okay, so I have this problem about a Senate committee with 5 Democrats, 5 Republicans, and 1 Independent. They need to sit around a circular table, and all the members of each party have to sit next to each other. I need to figure out in how many different ways this can happen. The problem also mentions that two seatings are considered the same if one is just a rotation of the other. Hmm, okay, let me break this down. First, since it's a circular table, the number of arrangements is usually less than a straight line because rotations are considered the same. <omitted> So, let me think. There are three parties: Democrats, Republicans, and Independents. Each party is a group that needs to be seated together. So, instead of thinking of 11 individuals, I can think of three blocks: D (Democrats), R (Republicans), and I (Independent). Each block has to be seated together, so the problem reduces to arranging these three blocks around the table. <omitted> So, for three blocks, it's $2! = 2$ distinct arrangements. But now, within each block, the members can be arranged among themselves. For the Democrats, there are 5 people, so they can be arranged in $5!$ ways. Similarly, the Republicans can be arranged in $5!$ ways. The Independent is just one person, so there's only 1 way to arrange that block. Therefore, the total number of arrangements is the number of ways to arrange the blocks multiplied by the number of ways to arrange the people within each block. So, that would be $2! * (5! * 5! * 1!) = 2 * (120 * 120 * 1) = 2 * 14400 = 28800$. </think> Early Exit ! [Conclusion] The Senate committee consists of three parties: Democrats (D), Republicans (R), and an Independent (I). Since they must sit together, we treat each party as a single block. 1. Arranging the Blocks: Around a circular table, the number of ways to arrange 3 blocks is $\frac{(3-1)!}{1} = 2!$. 2. Arranging Within Blocks: - Democrats can be arranged in $5!$ ways. - Republicans can be arranged in $5!$ ways. - The Independent is a single person, so there's $1!$ way. 3. Total Arrangements: Multiply the arrangements of the blocks by the arrangements within each block: $2! * 5! * 5! * 1! = 2 * 120 * 120 * 1 = 28,800$ ** Answer **: $\boxed{28800}$		[Slow thinking] Okay, so I have this problem about a Senate committee with 5 Democrats, 5 Republicans, and 1 Independent. They need to sit around a circular table, and all the members of each party have to sit next to each other. <omitted> Therefore, the total number of arrangements is the number of ways to arrange the blocks multiplied by the number of ways to arrange the people within each block. So, that would be $2! * (5! * 5! * 1!) = 2 * (120 * 120 * 1) = 2 * 14400 = 28800$. Wait, hold on. Is that all? Let me double-check. So, we have three blocks: D, R, I. Arrange them around the table: $(3-1)! = 2$. Then, within each block, arrange the members: $5!$ for D, $5!$ for R, and $1!$ for I. So, yes, $2 * 5! * 5! * 1! = 2 * 120 * 120 = 28800$. But wait, another thought: since the table is circular, does the direction matter? Like, clockwise vs. counterclockwise. In circular permutations, sometimes arrangements that are mirror images are considered different unless specified otherwise. <omitted> But let me think again. Is there another way to approach this problem? Maybe by considering the circular arrangement as a linear arrangement where we fix one person's position to eliminate rotational symmetry. <omitted> But since we've fixed the Democrats, the remaining seats are in a line, so the number of ways to arrange the Republican block and the Independent is $2!$ (since there are two blocks: R and I). Then, within the Republican block, they can be arranged in $5!$ ways, and the Independent is just 1. So, the total number of arrangements would be 1 (for the fixed Democrat) $* 2! * 5! * 1! = 2 * 120 = 240$. Wait, that's different from the previous answer. Hmm, so which one is correct? Wait, no, actually, if we fix one Democrat's seat, we're not just fixing that one seat, but the entire block of Democrats. So, the number of ways to arrange the blocks is $2!$ (for R and I), and within each block, it's $5!$ for R and $1!$ for I. So, $2! * 5! * 1! = 2 * 120 = 240$. But that seems too low because earlier we had 28800. Wait, so which approach is correct? Let me think. <omitted> Wait, so which approach is correct? Let me think. <omitted> Wait, so which approach is correct? Let me think.	
[No Conclusion] Token number: 870		Repetition Token number: 16384	

图 13: 在 MATH-500 上, DEER 与 Vanilla CoT 生成内容样本的对比。如绿色框所示, DEER 和 Vanilla CoT 都在第一个推理步骤中得出了正确答案。区别在于, DEER 在评估试验答案的置信度足够高后提前退出, 从而产生了正确结果。

Question: The coordinates of a parallelogram are (5, 3), (6, 8), (7, 4) and (x, y) and $x > 7$. What is the value of $x + y$?

Answer: **DEER**

Okay, so I have this problem where I need to find the coordinates of the fourth vertex of a parallelogram. The given points are (5, 3), (6, 8), (7, 4), and (x, y) with the condition that $x > 7$. Then, I need to find the value of $x + y$. Hmm, let me think about how to approach this. First, I remember that in a parallelogram, opposite sides are equal and parallel. That means the midpoints of the diagonals should be the same. So, if I can find the midpoint of one diagonal, it should be equal to the midpoint of the other diagonal. Let me label the points to make it clearer. Let's say point A is (5, 3), point B is (6, 8), point C is (7, 4), and point D is (x, y) . Now, I need to figure out which points are connected to each other. Since it's a parallelogram, the order of the points matters. I think the points should be connected in order, so A to B to C to D and back to A. But I'm not entirely sure, so maybe I should consider different possibilities.

Let me try another approach. Another property of a parallelogram is that the diagonals bisect each other. So, the midpoint of diagonal AC should be the same as the midpoint of diagonal BD. Let me calculate the midpoint of AC first. Point A is (5, 3) and point C is (7, 4). The midpoint formula is $(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2})$. So, midpoint of AC is $(\frac{5 + 7}{2}, \frac{3 + 4}{2}) = (12/2, 7/2) = (6, 3.5)$.

Midpoint is $(\frac{5 + x}{2}, \frac{3 + y}{2})$. Midpoint of BC: Point B is (6, 8) and point C is (7, 4). Midpoint is $(\frac{6 + 7}{2}, \frac{8 + 4}{2}) = (13/2, 12/2) = (6.5, 6)$. Setting midpoints equal: $\frac{5 + x}{2} = 6.5$ and $\frac{3 + y}{2} = 6$. Solving the first equation: $5 + x = 13 \Rightarrow x = 8$. Solving the second equation: $3 + y = 12 \Rightarrow y = 9$. Okay, this gives $x = 8$ and $y = 9$, which satisfies $x > 7$. So, the fourth point is (8, 9), and $x + y = 8 + 9 = 17$.

To find the coordinates of the fourth vertex of the parallelogram, we use the property that the diagonals of a parallelogram bisect each other. 1. Calculate the midpoint of diagonal BC: Points B (6, 8) and C (7, 4). Midpoint formula: $(\frac{6 + 7}{2}, \frac{8 + 4}{2}) = (13/2, 12/2) = (6.5, 6)$. 2. Set the midpoint of diagonal AD equal to the midpoint of BC: Let D be (x, y) . Midpoint of AD: $(\frac{5 + x}{2}, \frac{3 + y}{2})$. Equate to (6.5, 6): $\frac{5 + x}{2} = 6.5 \Rightarrow 5 + x = 13 \Rightarrow x = 8$. $\frac{3 + y}{2} = 6 \Rightarrow 3 + y = 12 \Rightarrow y = 9$. 3. Calculate $x + y$: $8 + 9 = 17$. Answer: **17**

Token number: ~800

Vanilla CoT

Okay, so I have this problem where I need to find the coordinates of the fourth vertex of a parallelogram. The given points are (5, 3), (6, 8), (7, 4), and (x, y) with the condition that $x > 7$. Then, I need to find the value of $x + y$. Hmm, let me think about how to approach this. First, I remember that in a parallelogram, opposite sides are equal and parallel. That means the midpoints of the diagonals should be the same. So, if I can find the midpoint of one diagonal, it should be equal to the midpoint of the other diagonal. Let me label the points to make it clearer. Let's say point A is (5, 3), point B is (6, 8), point C is (7, 4), and point D is (x, y) . Now, I need to figure out which points are connected to each other. Since it's a parallelogram, the order of the points matters. I think the points should be connected in order, so A to B to C to D and back to A. But I'm not entirely sure, so maybe I should consider different possibilities.

Let me try another approach. Another property of a parallelogram is that the diagonals bisect each other. So, the midpoint of diagonal AC should be the same as the midpoint of diagonal BD.

Midpoint is $(\frac{5 + x}{2}, \frac{3 + y}{2})$. Midpoint of BC: Point B is (6, 8) and point C is (7, 4). Midpoint is $(\frac{6 + 7}{2}, \frac{8 + 4}{2}) = (13/2, 12/2) = (6.5, 6)$. Setting midpoints equal: $\frac{5 + x}{2} = 6.5$ and $\frac{3 + y}{2} = 6$. Solving the first equation: $5 + x = 13 \Rightarrow x = 8$. Solving the second equation: $3 + y = 12 \Rightarrow y = 9$. Okay, this gives $x = 8$ and $y = 9$, which satisfies $x > 7$. So, the fourth point is (8, 9), and $x + y = 8 + 9 = 17$.

Wait, let me verify this. If the points are A(5,3), B(6,8), C(7,4), and D(8,9), does this form a parallelogram? Let me check the vectors.

These aren't equal either. Hmm, that's confusing. Wait, maybe I made a mistake in assigning the points. Let me try another approach.

Hmm, these aren't equal. Wait, that's a problem. Wait, but according to the vector addition, D should be (8,9). So, why are vectors AD and BC not equal? Wait, maybe I need to check the other pair of sides. Vector AD and vector BC are not equal, but vector AB and vector DC are equal.

These aren't equal, but in a parallelogram, both pairs should be equal. So, that's a problem. Wait, but according to the midpoint method, it's a parallelogram. So, maybe I'm missing something.

No answer !

Token number: 16384

图 14: 在 MATH-500 上, DEER 与 Vanilla 生成内容的对比。

Question: Point A lies somewhere within or on the square which has opposite corners at $(0,0)$ and $(2,2)$. Point B lies somewhere within or on the square which has opposite corners at points $(4,2)$ and $(5,3)$. What is the greatest possible value of the slope of the line containing points A and B ? Express your answer as a common fraction.

DEER

Okay, so I have this problem where I need to find the greatest possible slope of a line connecting two points, A and B. Point A is somewhere inside or on a square with opposite corners at $(0,0)$ and $(2,2)$. Point B is inside or on another square with opposite corners at $(4,2)$ and $(5,3)$. I need to figure out the maximum slope possible between any two points A and B in these squares.

First, let me visualize the squares. The first square has corners at $(0,0)$ and $(2,2)$. Similarly, the second square has opposite corners at $(4,2)$ and $(5,3)$. Hmm, so that square is a bit smaller. Let me figure out its sides. The distance between $(4,2)$ and $(5,3)$ is $\sqrt{(5-4)^2 + (3-2)^2} = \sqrt{1+1} = \sqrt{2}$. So, it's a square with side length $\sqrt{2}$, rotated 45 degrees relative to the axes.

Wait, actually, no. If the opposite corners are $(4,2)$ and $(5,3)$, then the sides are not aligned with the axes.

But wait, is that achievable? Because point A is in the first square, so if A is at $(2,0)$, which is a corner, and point B is at $(4,3)$, which is a corner of the second square. So, connecting $(2,0)$ to $(4,3)$ would give a slope of $(3-0)/(4-2) = 3/2$. But is that the maximum?

To find the greatest possible slope of the line containing points A and B , we need to maximize the expression $\frac{y_B - y_A}{x_B - x_A}$.

Point A lies within or on the square with opposite corners at $(0,0)$ and $(2,2)$. The minimum y -coordinate for A is 0 and the maximum x -coordinate is 2 . Point B lies within or on the square with opposite corners at $(4,2)$ and $(5,3)$. The maximum y -coordinate for B is 3 and the minimum x -coordinate is 4 .

To maximize the slope, we should choose A at $(2,0)$ and B at $(4,3)$. The slope is then $\frac{3-0}{4-2} = \frac{3}{2}$.

However, upon closer inspection, the square for B is actually a diamond with corners at $(4,2)$, $(5,3)$, $(5,2)$, and $(4,3)$. The point $(4,3)$ is indeed a corner of this square. Therefore, the maximum slope is achievable.

The greatest possible slope is $\frac{3}{2}$.

Token number: ~1000

Vanilla CoT

Okay, so I have this problem where I need to find the greatest possible slope of a line connecting two points, A and B. Point A is somewhere inside or on a square with opposite corners at $(0,0)$ and $(2,2)$. Point B is inside or on another square with opposite corners at $(4,2)$ and $(5,3)$. I need to figure out the maximum slope possible between any two points A and B in these squares.

First, let me visualize the squares. The first square has corners at $(0,0)$ and $(2,2)$. Similarly, the second square has opposite corners at $(4,2)$ and $(5,3)$. Hmm, so that square is a bit smaller. Let me figure out its sides. The distance between $(4,2)$ and $(5,3)$ is $\sqrt{(5-4)^2 + (3-2)^2} = \sqrt{1+1} = \sqrt{2}$. So, it's a square with side length $\sqrt{2}$, rotated 45 degrees relative to the axes.

Wait, actually, no. If the opposite corners are $(4,2)$ and $(5,3)$, then the sides are not aligned with the axes.

But wait, is that achievable? Because point A is in the first square, so if A is at $(2,0)$, which is a corner, and point B is at $(4,3)$, which is a corner of the second square. So, connecting $(2,0)$ to $(4,3)$ would give a slope of $(3-0)/(4-2) = 3/2$. But is that the maximum?

Wait, but maybe we can get a steeper slope by choosing points that are not at the corners. Because if we can have a larger $y_B - y_A$ with a smaller $x_B - x_A$, that would give a steeper slope.

Wait, but hold on. Let me think again. Maybe if we don't take A at $(2,0)$ and B at $(4,3)$, but somewhere else, we can get a steeper slope.

But wait, is there a way to have a larger slope? For example, if A is at $(2,2)$ and B is at $(4,3)$. Then, the slope is $(3-2)/(4-2) = 1/2$, which is less than $3/2$.

Alternatively, if A is at $(0,0)$ and B is at $(5,3)$, then the slope is $(3-0)/(5-0) = 3/5$, which is less than $3/2$.

Alternatively, if point A is at (x, y_A) where $x < 2$, then point B would have to be at $(x+2, y_B)$. But since the second square starts at $x=4$, $x+2$ must be at least 4, so x must be at least 2.

No answer !

Token number: 16384

图 15: 在 MATH-500 上, DEER 与 Vanilla 生成内容的对比。

The expression $1 + 2 + 3 - 4 + 5 + 6$ is equivalent to 13. If we add parentheses in various locations, we can change this value. For example, $1+2+3-(4+5)+6=1+2+3-9+6=6-9+6=3$. If we only add parentheses (and not rearrange the order of the terms), what is the minimum possible value this expression can obtain? (For the purposes of this problem, we cannot change addition or subtraction into multiplication. For example, $(1+2+3)(-4)+5+6$ is not a valid placement of parentheses.) Enter your answer as a single value.

Answer:

DEER

Okay, so I have this math problem here: the expression is $1 + 2 + 3 - 4 + 5 + 6$, and it's equal to 13. The task is to add parentheses in various places to change the value, but I can't rearrange the terms or turn the operations into multiplication. I need to find the minimum possible value this expression can obtain by adding parentheses. Hmm, interesting.

First, let me make sure I understand the problem correctly. I can only add parentheses, which will change the order in which operations are performed, but I can't move the numbers around or change the plus and minus signs into multiplication or division. So, for example, I can do something like $(1 + 2) + 3 - 4 + 5 + 6$, but I can't do $1*(2+3) - 4 + 5 + 6$ because that would involve multiplication, which isn't allowed.

<omitted>

Yep, that's correct. So, 13 is the original value.

Alternatively, group $(1 + 2) + (3 - 4 + 5 + 6)$. So, $1 + 2 = 3$, $3 - 4 = -1$, $-1 + 5 = 4$, $4 + 6 = 10$. Then, $3 + 10 = 13$.

Alternatively, group $1 + (2 + 3 - 4 + 5) + 6$. So, $2 + 3 = 5$, $5 - 4 = 1$, $1 + 5 = 6$. Then, $1 + 6 + 6 = 13$.

Hmm, seems like 3 is the lowest I can get so far. But let me think if there's another way.

What if I group $1 + 2 + 3 - (4 + 5 + 6)$. Let's compute that: $4 + 5 + 6 = 15$. Then, $1 + 2 + 3 = 6$, $6 - 15 = -9$. So, that's -9. Wait, that's way lower than 3. Is that allowed?

</think>

To find the minimum possible value of the expression $1 + 2 + 3 - 4 + 5 + 6$ by adding parentheses, we can strategically group terms to maximize the impact of the subtraction.

By grouping the terms after the subtraction, we can create a larger number to subtract, significantly reducing the overall value. For example, grouping $(4 + 5 + 6)$ together:

$$1 + 2 + 3 - (4 + 5 + 6) = 6 - 15 = -9$$

Thus, the minimum possible value is $\boxed{-9}$.

Token number: ~1500

Vanilla CoT

Okay, so I have this math problem here: the expression is $1 + 2 + 3 - 4 + 5 + 6$, and it's equal to 13. The task is to add parentheses in various places to change the value, but I can't rearrange the terms or turn the operations into multiplication. I need to find the minimum possible value this expression can obtain by adding parentheses. Hmm, interesting.

First, let me make sure I understand the problem correctly. I can only add parentheses, which will change the order in which operations are performed, but I can't move the numbers around or change the plus and minus signs into multiplication or division. So, for example, I can do something like $(1 + 2) + 3 - 4 + 5 + 6$, but I can't do $1*(2+3) - 4 + 5 + 6$ because that would involve multiplication, which isn't allowed.

<omitted>

Yep, that's correct. So, 13 is the original value.

Alternatively, group $(1 + 2) + (3 - 4 + 5 + 6)$. So, $1 + 2 = 3$, $3 - 4 = -1$, $-1 + 5 = 4$, $4 + 6 = 10$. Then, $3 + 10 = 13$.

Alternatively, group $1 + (2 + 3 - 4 + 5) + 6$. So, $2 + 3 = 5$, $5 - 4 = 1$, $1 + 5 = 6$. Then, $1 + 6 + 6 = 13$.

Hmm, seems like 3 is the lowest I can get so far. But let me think if there's another way.

What if I group $1 + 2 + 3 - (4 + 5 + 6)$. Let's compute that: $4 + 5 + 6 = 15$. Then, $1 + 2 + 3 = 6$, $6 - 15 = -9$. So, that's -9. Wait, that's way lower than 3. Is that allowed?

Wait, hold on. The original expression is $1 + 2 + 3 - 4 + 5 + 6$. If I group $4 + 5 + 6$, that would be subtracting 15, so $6 - 15 = -9$.

<omitted>

Wait, but in the original expression, after the subtraction, it's $+5 + 6$. So, if I group $4 + 5 + 6$, that would be changing the operations after the subtraction, but I can't rearrange terms. So, is that allowed?

<omitted>

Alternatively, group $1 + (2 + 3 - 4 + 5 + 6)$. As before, $1 + 12 = 13$.

Alternatively, group $(1 + 2 + 3 - 4 + 5) + 6$

Endless exhaustive search!

Token number: 16384

图 16: 在 MATH-500 上, DEER 与 Vanilla 生成内容的对比。

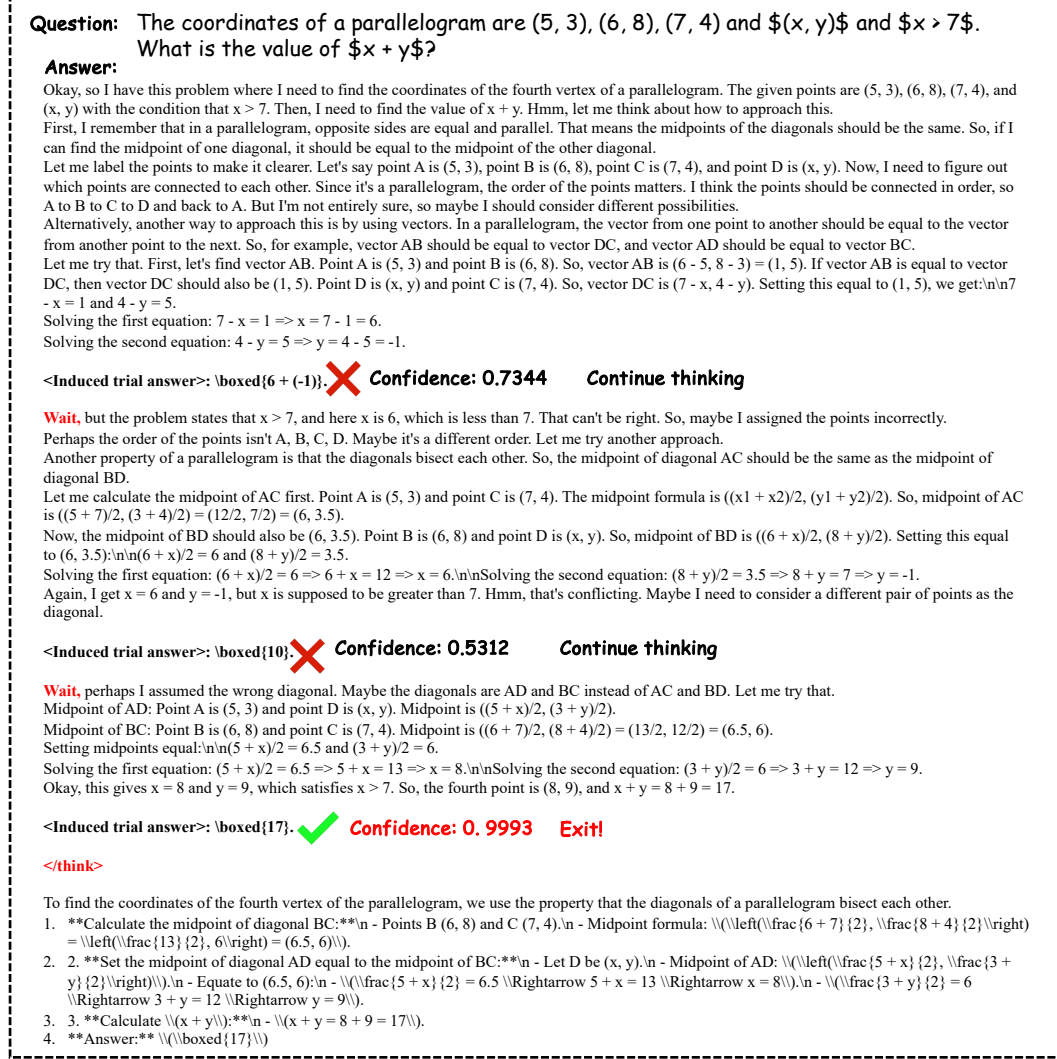


图 17: 一个展示 LRM 感知珍珠推理涌现的隐含能力的示例。当模型的推理内容不足以支持引出的答案时，置信度保持在较低水平，如图中前两个置信度值所示。相反，当模型提供了充分且合理的推理作为依据时，置信度达到高水平，可能接近 1，如图中最后一个置信度值所示。

Question: Define $p = \sum_{k=1}^{\infty} \frac{1}{k^2}$ and $q = \sum_{k=1}^{\infty} \frac{1}{k^3}$. Find a way to write $\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j+k)^3}$ in terms of p and q .

Answer:

DeepSeek-R1-Distill-Qwen-32B

Length

Okay, so I have this problem where I need to express the double sum $\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j+k)^3}$ in terms of p and q , where $p = \sum_{k=1}^{\infty} \frac{1}{k^2}$ and $q = \sum_{k=1}^{\infty} \frac{1}{k^3}$. Hmm, let me think about how to approach this. First, I know that $\sum_{k=1}^{\infty} \frac{1}{k^2}$ is the famous Basel problem, which equals $\frac{\pi^2}{6}$, and $\sum_{k=1}^{\infty} \frac{1}{k^3}$ is the Riemann zeta function at 3, which is known as Apery's constant, approximately 1.202... But I don't think I need the exact values; instead, I need to relate the double sum to these series. Looking at the double sum, it's over all positive integers j and k , and the term is $\frac{1}{(j+k)^3}$. Maybe I can change variables to simplify this. Let me set $n = j + k$. Then, for each fixed n , j can range from 1 to $n-1$, and $k = n - j$. So, perhaps I can rewrite the double sum as a single sum over n , where each term is the sum over j from 1 to $n-1$ of $\frac{1}{n^3}$. Let me write that down: $\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j+k)^3} = \sum_{n=2}^{\infty} \sum_{j=1}^{n-1} \frac{1}{n^3}$. Yes, that makes sense because the smallest n can be is 2 (when $j=1$ and $k=1$), and it goes to infinity. Now, for each n , the inner sum is just adding $\frac{1}{n^3}$ a total of $(n-1)$ times. So, the inner sum simplifies to $\frac{(n-1)}{n^3}$. Now, the double sum becomes $\sum_{n=2}^{\infty} \frac{(n-1)}{n^3}$. Hmm, that seems manageable. Let me write this as: $\sum_{n=2}^{\infty} \left(\frac{n}{n^3} - \frac{1}{n^3} \right) = \sum_{n=2}^{\infty} \left(\frac{1}{n^2} - \frac{1}{n^3} \right)$. Yes, that's correct. So, this is equal to $\sum_{n=2}^{\infty} \frac{1}{n^2} - \sum_{n=2}^{\infty} \frac{1}{n^3}$. But I know that $\sum_{n=1}^{\infty} \frac{1}{n^2} = p$ and $\sum_{n=1}^{\infty} \frac{1}{n^3} = q$. So, if I subtract the $n=1$ terms from both series, I can express the sums starting from $n=2$. Let's compute that: $\sum_{n=2}^{\infty} \frac{1}{n^2} = p - \frac{1}{1^2} = p - 1$. Similarly, $\sum_{n=2}^{\infty} \frac{1}{n^3} = q - \frac{1}{1^3} = q - 1$. So, substituting back into our expression: $\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j+k)^3} = (p - 1) - (q - 1) = p - q$.
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So, the double sum simplifies to $p - q$. Therefore, the answer is $p - q$.

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QwQ-32B

Okay, so I need to find a way to express the double sum $\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j+k)^3}$ in terms of p and q , where p is the sum of $\frac{1}{k^2}$ and q is the sum of $\frac{1}{k^3}$. Hmm, let me think about how to approach this. First, I remember that double sums can sometimes be simplified by changing variables or switching the order of summation. Maybe I can let $n = j + k$, so that for each n , the terms where $j + k = n$ contribute to the sum. That might help me rewrite the double sum as a single sum over n , with each term multiplied by the number of pairs (j, k) that add up to n . Let me try that. Let's set $n = j + k$. Then for each $n \geq 2$ (since j and k are at least 1), the number of pairs (j, k) such that $j + k = n$ is $(n-1)$. For example, when $n=2$, there's only one pair (1,1); when $n=3$, there are two pairs (1,2) and (2,1), and so on. So the double sum can be rewritten as $\sum_{n=2}^{\infty} \frac{(n-1)}{n^3}$. Wait, is that right? Let me check. Each term $\frac{1}{(j+k)^3}$ for each pair (j, k) would be grouped by $n = j + k$, so each $(n-1)$ terms each of $\frac{1}{n^3}$. Therefore, the total contribution for each n is $\frac{(n-1)}{n^3}$. So the entire double sum becomes $\sum_{n=2}^{\infty} \frac{(n-1)}{n^3}$. Hmm, okay. Now I need to express this in terms of p and q . Let me see. Let's first write out the sum: $\sum_{n=2}^{\infty} \left(\frac{n}{n^3} - \frac{1}{n^3} \right) = \sum_{n=2}^{\infty} \left(\frac{1}{n^2} - \frac{1}{n^3} \right)$. Ah, that simplifies to the difference between two sums. Let's split that into two separate sums: $\sum_{n=2}^{\infty} \frac{1}{n^2} - \sum_{n=2}^{\infty} \frac{1}{n^3}$. Now, notice that these are just the tails of the series for p and q . Since $p = \sum_{k=1}^{\infty} \frac{1}{k^2}$, then $\sum_{n=2}^{\infty} \frac{1}{n^2} = p - \frac{1}{1^2} = p - 1$. Similarly, $q = \sum_{k=1}^{\infty} \frac{1}{k^3}$, so $\sum_{n=2}^{\infty} \frac{1}{n^3} = q - \frac{1}{1^3} = q - 1$. Therefore, substituting back into the expression: $\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j+k)^3} = (p - 1) - (q - 1) = p - q$.
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Let's summarize the above thought process and provide the result. The double sum can be rewritten by considering the sum over $n = j + k$. For each $n \geq 2$, there are $(n-1)$ pairs (j, k) such that $j + k = n$. This transforms the double sum into a single sum: $\sum_{n=2}^{\infty} \frac{(n-1)}{n^3}$. Breaking this into two separate sums: $\sum_{n=2}^{\infty} \left(\frac{n}{n^3} - \frac{1}{n^3} \right) = \sum_{n=2}^{\infty} \left(\frac{1}{n^2} - \frac{1}{n^3} \right)$. Recognizing these as the tails of the series for p and q , we subtract the first terms: $\sum_{n=2}^{\infty} \frac{1}{n^2} = p - 1$ and $\sum_{n=2}^{\infty} \frac{1}{n^3} = q - 1$. Thus, the double sum simplifies to $(p - 1) - (q - 1) = p - q$.
 Answer: $p - q$.
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To express the double sum $\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j+k)^3}$ in terms of p and q , we start by changing variables. Let $n = j + k$. For each $n \geq 2$, the number of pairs (j, k) such that $j + k = n$ is $(n-1)$. This allows us to rewrite the double sum as $\sum_{n=2}^{\infty} \frac{(n-1)}{n^3}$. Next, we split this into two separate sums: $\sum_{n=2}^{\infty} \left(\frac{n}{n^3} - \frac{1}{n^3} \right) = \sum_{n=2}^{\infty} \left(\frac{1}{n^2} - \frac{1}{n^3} \right)$. Recognizing these as the tails of the series for p and q , we have $\sum_{n=2}^{\infty} \frac{1}{n^2} = p - 1$ and $\sum_{n=2}^{\infty} \frac{1}{n^3} = q - 1$. Substituting these into the expression gives: $\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j+k)^3} = (p - 1) - (q - 1) = p - q$.
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图 18: 一个展示 QwQ-32B 中存在多个 `</think>` 标记的示例。对于两个模型, 绿色思考块都足以得出正确结论。然而, QwQ-32B 继续进行了一个额外的总结 (红色块) 并生成了自己的 `</think>` 标记。基于以上所有内容, 它得出了结论。